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## COMPOSITIONS AND METHODS FOR TREATMENT OF KIDNEY DISEASE

### Abstract

Described herein are antisense oligonucleotides, vectors, and related compositions and methods for increasing expression of PKD1 mRNA and Polycystin 1 protein and uses thereof for the treatment of autosomal dominant polycystic kidney disease (ADPKD).

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## **Background/Summary**

[0001] The present application claims priority from AU 2023901907 filed on 16 Jun. 2023, and AU 2023903042 filed on 21 Sep. 2023, the entire contents of each of which are incorporated herein by reference.

### **TECHNICAL FIELD**

[0002] The present disclosure generally is directed to oligonucleotides and related compositions and methods for treating conditions associated with mutations in the Polycystic Kidney Disease 1 (PKD1) gene.

### **BACKGROUND**

[0003] Autosomal dominant polycystic kidney disease (ADPKD) is the most common inherited nephropathy that leads to end-stage renal disease (ESRD). Approximately 1 in 500-2,500 individuals carry a mutation for this condition. ADPKD is a progressive disorder that is characterised by an abnormal expansion of renal tubule cells resulting in the growth of multiple cysts in the kidney.

[0004] Cyst development starts early in life and in the more severe cases in utero and is commonly followed by a prolonged period of asymptomatic progression. Signs and symptoms of ADPKD usually present between the ages of 30 and 40. However, approximately 3% of ADPKD patients have either very-early-onset or unusually rapid progressive disease. Consistent with the decline in renal function patients present with various urinary complications such as cyst and urinary tract infections, a decline in glomerular filtration rate, chronic lower back pain and hypertension.

ADPKD patients often present with additional extra-renal conditions including hepatic cysts (in >90% of patients aged >35 years), pancreatic cysts, intracranial aneurysms, colon diverticulosis, and heart valve defects. Overall, ADPKD is associated with significant morbidity a reduced life expectancy.

[0005] ADPKD is predominantly caused by a mutation in one of the polycystin genes; PKD1 (Polycystin 1, Transient Receptor Potential Channel Interacting) (74-85% of patients) or PKD2 (15-26%). The phenotype of patients with a PKD1 mutation is usually more severe compared to those with PKD2 mutations, which is reflected in approximately 20 years difference in the mean age of ESRD (54.3 years in PKD1 vs 74.0 years in PKD2 disease).

[0006] There is an ongoing need to provide new treatments or preventative measures for ADPKD.

### **SUMMARY**

[0007] PKD1 consists of 46 exons spanning approximately 50 kb of genomic DNA on the short arm of chromosome 16 (16p13.3). The canonical PKD1-encoded Polycystin 1 protein is a 4,303 amino acid, glycosylated integral membrane protein that localises to primary cilia, endoplasmic reticulum, adherent and desmosomal junctions, apical membranes, plasma membrane and junctional complexes. Polycystin 1 contains a large N-terminal extracellular region, multiple transmembrane domains, and a cytoplasmic C-tail and functions as a regulator of calcium-permeable cation channels and intracellular calcium homeostasis. It is also involved in cell-cell/matrix interactions and modulation of G-protein-coupled signal-transduction pathways. Splice variants encoding different isoforms have been noted for this gene.

[0008] While not wishing to be bound by theory, it is believed that kidney cysts develop in ADPKD once the level of Polycystin 1 function falls below a specific level. Indeed, this is consistent with the fact that the level of residual Polycystin 1 function from the mutated gene directly correlates with disease severity. The median age at onset of ESRD was 55 years for carriers of a truncating mutation (complete loss of function) and 67 years for carriers of a non-truncating mutation (partial loss of function).

[0009] The present disclosure provides antisense oligonucleotides (ASO), antisense RNA (AR) expression vectors, and related compositions and methods to increase PKD1 mRNA and/or Polycystin 1 protein levels by modulating the post-transcriptional or translational regulation of the 3' untranslated region (UTR) of PKD1 mRNA, e.g., by blocking specific binding of mirR-17 family miRNAs (e.g., miR-17-5p, miR-106a-5p, miR-106b-5p, miR-20a-5p, miR-93-5p) or miR-200 family miRNAs (e.g., miR-200b, miR200c, or miR-429) to their cognate binding sites.

[0010] Accordingly, in one aspect provided herein is an ASO that binds to a targeted portion of the 3' UTR of a PKD1 mRNA, wherein binding of the antisense oligonucleotide to the targeted portion increases the level of Polycystin 1 protein. In another aspect provided herein is an ASO that binds to a targeted portion of the 3' untranslated region (UTR) of a Polycystic Kidney Disease 1 (PKD1) mRNA, wherein binding of the antisense oligonucleotide to the targeted portion reduces kidney cyst growth or size when introduced into a 3D kidney cyst culture. In some examples binding of the antisense oligonucleotide to the targeted portion reduces specific binding of miR-17 family miRNAs (e.g., miR-17-5p, miR-106a-5p, miR-106b-5p, miR-20a-5p, miR-93-5p) or miR-200 family miRNAs (e.g., miR-200b, miR200c, or miR-429) to the 3' UTR. In some examples, where binding of the antisense oligonucleotide to the targeted portion reduces specific binding of mirR-17 family miRNAs (e.g., miR-17-5p, miR-106a-5p, miR-106b-5p, miR-20a-5p, miR-93-5p) or miR-200 family miRNAs (e.g., miR-200b, miR200c, or miR-429) to the 3' UTR, the antisense oligonucleotide comprises the sequence of any one of SEQ ID NOs:2-351 or 353-362.

[0011] In a further aspect provided herein is a vector for expression, in a mammalian cell, of an AR that binds to a targeted portion of the 3' UTR of a PKD1 mRNA, wherein binding of the AR to the targeted portion increases the level of Polycystin protein. In another aspect provided herein is a vector for expression, in a mammalian cell, of an AR that binds to a targeted portion of the 3' UTR of a PKD1 mRNA, wherein binding of the AR to the targeted portion reduces cyst growth when introduced into a 3D kidney cyst culture. In some examples binding of the AR to the targeted portion reduces specific binding of mirR-17 family miRNAs (e.g., miR-17-5p, miR-106a-5p, miR-106b-5p, miR-20a-5p, miR-93-5p) or miR-200 family miRNAs (e.g., miR-200b, miR200c, or miR-429) to the 3' UTR. In some examples, where binding of the AR to the targeted portion reduces specific binding of mirR-17 family miRNAs (e.g., miR-17-5p, miR-106a-5p, miR-106b-5p, miR-20a-5p, miR-93-5p) or miR-200 family miRNAs (e.g., miR-200b, miR200c, or miR-429) to the 3' UTR, the AR comprises or consists of the sequence of any one of any one of SEQ ID NOs:2-351 or 353-362. In some examples the vector is a non-viral vector. In other examples the vector is a viral vector. In some examples, where the vector is a viral vector, the viral vector is provided in a recombinant virus selected from the group consisting of: adeno-associated virus (AAV), adenovirus, lentivirus, and anellovirus.

[0012] In some examples the AR can be delivered by a vector (e.g., a plasmid or a recombinant virus) that comprises a kidney cell type-selective or tissue-selective promoter for driving expression of the AR in the mammalian cell. In some examples, the promoter is selective for expression in kidney cells selected from the group consisting of: pericytes, podocytes, parietal epithelial cells, proximal tubule cells, ascending loop of Henle cells, descending loop of Henle cells, distal tubule cells, connecting tubule cells, intercalated cells, principal cells, peritubular capillary endothelium cells, and glomerular endothelium cells. In some examples the vector includes an inducible promoter.

[0013] In some examples, in any of the foregoing ASOs or vectors, the ASO or the AR binds within

a targeted portion of the 3' UTR corresponding to SEQ ID NO:1. In some examples the nucleotide sequence of the ASO or AR is at least 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 100% complementary to the nucleotide sequence of the targeted portion over the length of the ASO or the AR. In some examples the nucleotide sequence of the ASO or AR comprises or consists of any one of SEQ ID NOs:2-351 or 353-362. In some examples the nucleotide sequence of the ASO or AR comprises or consists of any one of SEQ ID NOs: 17, 47, 73, 334-337, or 355-360. In some examples the nucleotide sequence of the ASO or AR comprises or consists of any one of SEQ ID NOs:334 or 335. In some examples the nucleotide sequence of the ASO or AR comprises or consists of any one of SEQ ID NOs:355-360. In some examples the nucleotide sequence of the ASO or AR comprises or consists of any one of SEQ ID NOs:17, 73, 336, and 337. In some examples the nucleotide sequence of the ASO or AR comprises or consists of SEQ ID NO:47.

[0014] In some examples any of the foregoing ASOs include a backbone modification. In some examples the backbone modification includes a phosphorothioate linkage or a phosphorodiamidate linkage. In other examples the ASO includes a phosphorodiamidate morpholino, a locked nucleic acid, a peptide nucleic acid, or a 2'-O-modification such as a 2'-O-methyl, a 2'-Fluoro, or a 2'-O-methoxyethyl moiety. In some examples the ASO includes at least one modified sugar moiety. In other examples each sugar moiety in the ASO is a modified sugar moiety. In some examples the ASO includes a 2'-O-methoxyethyl moiety. In other examples each nucleotide of the ASO includes a 2'-O-methoxyethyl moiety.

[0015] In some examples of any of the foregoing ASOs or vectors, the nucleotide sequence of the ASO or AR consists of 20 to 30 nucleotides, 22 to 30 nucleotides, 24 to 30 nucleotides, 25 to 30 nucleotides, or 26 to 30 nucleotides. In some examples the nucleotide sequence of the ASO or the AR consists of 25 to 30 nucleotides. In some examples, where the sequence of the ASO consists of 20 to 30 nucleotides, the ASO includes one or more phosphorodiamidate morpholino moieties.

[0016] In some examples any of the foregoing ASOs also includes a linked functional moiety. In some examples the functional moiety includes a delivery moiety. In some examples the delivery moiety is selected from the group consisting of: lipids, peptides, carbohydrates, polyethers (e.g., polyethylene glycol), and antibodies. In some examples, wherein the ASO includes a delivery moiety, the delivery moiety includes a cell-penetrating peptide (CPP). In some examples the delivery moiety includes a receptor binding domain (RBD). In some examples the delivery moiety includes a poly(ethylene glycol) (PEG) moiety. In some examples the delivery moiety includes a N-acetylgalactosamine (GalNAc) moiety. In some examples the delivery moiety includes a fatty acid or lipid moiety. In some embodiments the fatty acid chain length is about C8 to C20. In other examples the functional moiety includes a stabilising moiety. In some examples the functional moiety is covalently linked to the ASO. In other examples the functional moiety is non-covalently linked to the ASO. In some examples the functional moiety is linked to the 5' end of the ASO. In other examples the functional moiety is linked to the 3' end of the ASO.

[0017] In a related aspect provided herein is a pharmaceutical composition that includes any of the foregoing ASOs or vectors and a pharmaceutically acceptable excipient.

[0018] In a further related aspect provided herein is a method for treating autosomal ADPKD where a subject in need thereof is administered a therapeutically effective amount of the foregoing pharmaceutical composition. In some examples the subject to be treated is a human subject. In another related aspect provided herein is the use of any of the foregoing ASOs, vectors in the manufacture of a medicament for treating ADPKD.

[0019] In another aspect provided herein is a method for increasing PKD1 mRNA and/or Polycystin 1 protein in a cell ex vivo or in a tissue in vivo, where the method includes a step of contacting the cell or tissue with any of the foregoing ASOs, vectors, or pharmaceutical compositions.

[0020] In yet another aspect provided herein is a genetically modified cell comprising any of the

foregoing ASOs or vectors. In some examples the genetically modified cells are mammalian cells. In some examples the genetically modified mammalian cell is a genetically modified human cell.

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## Description

### BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS

[0021] FIG. 1—Schematic illustration of PKD1 exon-intron structure, miRNA binding sites in PKD 3' UTR, and relative binding positions of exemplary PMOs. (A) Exon/-Intron map of human PKD1 pre-mRNA; (B) sequence of PKD1 mRNA 3' UTR that includes binding site (bold) for miR-17 family miRNAs. The relative positions of complementary PMOs having sequences corresponding to SEQ ID NOs:47 (+604+628), 334 (+611+635), and 335 (+615+639), respectively, are indicated; (C) sequence of PKD1 mRNA 3' UTR including binding sites (bold) for miR-200 family miRNAs. The relative positions of complementary PMOs having sequences corresponding to SEQ ID NOs:14 (+505+529), 337 (+515+539), 336 (+677+701), 73 (+682+706) and 75 (+688+712), respectively, are indicated.

[0022] FIG. 2—Screening of PPMOs that target the miR-17 binding site in PKD1 in HEK293 cells.

[0023] Screening of PPMOs with sequences corresponding to SEQ ID NOs:47 (+604+628), 334 (+611+635), and 335 (+615+639) in HEK293 cells. The graph shows PKD1 mRNA expression at 24 h post treatment with miR-17 site targeting PPMOs or non-target control (GTC CTR). The PKD1 mRNA expression was normalized to TBP and expression in untreated cells was set to 1.

[0024] FIG. 3—Effect of PPMOs that target the sequence in the 3' UTR of PKD1 to which the miR-17 miRNA seed sequence binds in an ADPKD patient cell line

[0025] PPMOs that demonstrated significant PKD1 mRNA upregulation, as per Example 2, were selected for their ability to upregulate Polycystin 1 (PC1) protein in an ADPKD patient cell line carrying the PKD1 heterozygous mutation (p.Q2556\*). The graph shows the median fluorescence intensity (MFI) of PC1 protein staining on the cell surface, measured via flow cytometry 5 days post treatment with 10  $\mu$ M miR-17 site targeting PPMOs. The Polycystin protein staining MFI was normalized to Isotype stained antibody controls and the MFI in untreated cells was set to 1.

[0026] FIG. 4—Functional validation of PPMOs in a patient-derived 3D cyst model.

[0027] Functional validation of peptide PMOs (PPMOs) with sequences corresponding to SEQ ID NOs:47 (+604+628), 334 (+611+635), and 335 (+615+639) in patient derived cyst cells. The PPMOs were dosed at 1  $\mu$ M, 3  $\mu$ M, 10  $\mu$ M and 20  $\mu$ M and after 7 days of exposure, cultures are fixed and stained for actin cytoskeleton and nuclei. Growth and swelling of cysts are visualized by high-content microscopy imaging. Representative images show the cyst size in wells treated with 20  $\mu$ M PPMO 7 days post treatment. The assay was performed under unstimulated condition (spontaneous cyst formation).

[0028] FIG. 5—Quantification of patient-derived 3D cyst area and cell death

[0029] Images from all assay conditions described in FIG. 4 were further analysed to determine the dose-dependent change of cyst area ( $\mu$ m.sup.2) and cell death (%) to distinguish the efficacy and cytotoxicity. The cut-off for cell death was set at 15%.

[0030] (A-C) Swelling Inhibition was determined from measurements of Cyst Area. This was calculated as the average area of each cyst in every plane of the z-stack. (D-F) Cytotoxicity was calculated as the proportion of dead cells. Cells were scored as 'dead' if their nucleus was not associated with a colocalizing actin cytoskeleton (rhodamine-phalloidin labelled). This value was normalized to the solvent control (0%) and presented as mean of replicates $\pm$ standard deviation.

[0031] FIG. 6 Screening of PPMO microwalks for inhibition of miR-17 binding

[0032] PPMOs corresponding to oligo sequences in SEQ ID NOs:353-361, were incubated with HEK293 cells at concentrations of 3  $\mu$ M, 10  $\mu$ M, and 30  $\mu$ M for 24 hours, with n=3 technical replicates per treatment condition. A non-targeting control PPMO, designed not to hybridize to any

known human transcripts, was used as a negative control treatment. This control PPMO was conjugated to the same cell-penetrating peptide as the test PPMOs. A positive control oligonucleotide (RGLS4326, Med Chem Express, Catalogue No. HY-139290), an inhibitor of miR-17, was also included as an assay control. Data were normalized to two housekeeping genes, reported as relative levels compared to untreated cells, which were set to 1. Data represent mean±S.D. and UT=untreated cells. n=1 biological replicates. TBP=TATA-Binding Protein, DHX57=DExH-Box Helicase 57.

[0033] FIG. 7: Screening of PPMOs for inhibition of miR-200 binding

[0034] PPMOs corresponding to oligonucleotide sequences in SEQ ID NOs:14, 73, 75, 336-337 were incubated with HEK293 cells at concentrations of 3 μM, 10 μM, and 30 μM for 24 hours, with n=3 technical replicates per treatment condition. A non-targeting control PPMO, designed not to hybridize to any known human transcripts, was used as a negative control treatment. This control PPMO was conjugated to the same cell-penetrating peptide as the test PPMOs. A positive control oligo (RGLS4326, Med Chem Express, Catalogue No. HY-139290), an inhibitor of miR-17, was also included. Data were normalized to two housekeeping genes, reported as relative levels compared to untreated cells, which were set to 1. Data represent mean±S.D. and UT=untreated cells. n=2 biological replicates. TBP=TATA-Binding Protein, DHX57=DExH-Box Helicase 57.

[0035] FIG. 8: Western blot images of PPMO-treated HEK293 cells

[0036] HEK293 cells were treated with PPMOs (+604+628) and (+611+635) with the sequences corresponding to SEQ ID NOs:47, and 334, respectively. A non-targeting control (NTC) was used as a negative control while an inhibitor of miR-17 (RGLS4326, Med Chem Express, Catalogue No. HY-139290), was included as a positive control. PPMO-treated cells were harvested at 5 days and analysed for PC1 expression using western blot assay. Two protein bands at approximately 462 and 350 kDa indicate full-length (FL) and N-terminal fragment (NTF) of PC1, respectively. The intensity of total protein stain was analysed as loading control. Experiment was performed in quadruplicate.

[0037] FIG. 9: Quantification of PC1 protein upregulation in PPMO-treated HEK293 cells

[0038] Bar graph represents mean±S.D. of PC1 protein expression analysed using gel images in FIG. 8. Both FL and NTF bands were included in the analysis and normalized to total protein staining. A dash line indicates baseline PC1 protein level of untreated which was set as 1. Experimental was performed in quadruplicate. n=2 biological replicates. UT=untreated cells. NTC=non-targeting control.

[0039] FIG. 10: PC1 protein analysis in PPMO-treated WT9-7 cells

[0040] WT9-7 cells were incubated with PPMOs with the sequences corresponding to SEQ ID NO:47 (+604+628) in quadruplicate and a non-targeting control (NTC) for 2, 3 or 5 days. A non-targeting control (NTC) was used as a negative control and an inhibitor of miR-17 (RGLS4326, Med Chem Express, Catalogue number HY-139290), was included as a positive control. Bar graph represents mean±S.D. of PC1 protein normalized to total protein stain in relative to untreated cells. n=1 biological replicate. UT=untreated cells. NTC=non-targeting control.

## DETAILED DESCRIPTION

### General

[0041] Throughout this specification, unless specifically stated otherwise or the context requires otherwise, reference to a single step, composition of matter, group of steps or group of compositions of matter shall be taken to encompass one and a plurality (i.e., one or more) of those steps, compositions of matter, groups of steps or groups of compositions of matter. Thus, as used herein, the singular forms “a”, “an” and “the” include plural aspects unless the context clearly dictates otherwise. For example, reference to “a” includes a single as well as two or more; reference to “an” includes a single as well as two or more; reference to “the” includes a single as well as two or more and so forth.

[0042] Each example of the present disclosure described herein is to be applied mutatis mutandis to

each and every other example unless specifically stated otherwise.

[0043] Those skilled in the art will appreciate that the disclosure herein is susceptible to variations and modifications other than those specifically described. It is to be understood that the disclosure includes all such variations and modifications. The disclosure also includes all of the steps, features, compositions and compounds referred to or indicated in this specification, individually or collectively, and any and all combinations or any two or more of said steps or features.

[0044] The present disclosure is not to be limited in scope by the specific examples described herein, which are intended for the purpose of exemplification only. Functionally-equivalent products, compositions and methods are clearly within the scope of the disclosure, as described herein.

[0045] The present disclosure is performed without undue experimentation using, unless otherwise indicated, conventional techniques of molecular biology, microbiology, virology, recombinant DNA technology, peptide synthesis in solution, solid phase peptide synthesis, and immunology. Such techniques are described and explained throughout the literature in sources such as Perbal 1984, Sambrook et al., 2001, Brown (editor) 1991, Glover and Hames (editors) 1995 and 1996, Ausubel et al. including all updates until present, Coligan et al. (editors) (including all updates until present), Maniatis et al. 1982, Gait (editor) 1984, Hames and Higgins (editors) 1984, Freshney (editor) 1986.

[0046] The term “and/or”, e.g. “X and/or Y” shall be understood to mean either “X and Y” or “X or Y” and shall be taken to provide explicit support for both meanings or for either meaning.

[0047] The term “about”, unless stated to the contrary, refers to  $\pm 20\%$ , more preferably  $\pm 10\%$ , of the designated value. For the avoidance of doubt, the term “about” followed by a designated value is to be interpreted as also encompassing the exact designated value itself (for example, “about 10” also encompasses 10 exactly).

[0048] Throughout this specification the word “comprise”, or variations such as “comprises” or “comprising”, will be understood to imply the inclusion of a stated element, integer or step, or group of elements, integers or steps, but not the exclusion of any other element, integer or step, or group of elements, integers or steps.

[0049] The term “antisense oligonucleotide” “antisense oligomer” or “ASO,” as used herein, encompasses oligonucleotides and any other oligomeric molecule that comprises nucleobases capable of hybridizing to a complementary sequence on a target RNA transcript, including, but not limited to, those that do not comprise a sugar moiety, such as in the case of a peptide nucleic acid (PNA). Preferably, the ASO is an ASO that is resistant to nuclease cleavage or degradation.

[0050] The phrase “binds to a targeted portion” or “binds within a targeted portion,” in reference to an ASO or AR, as used herein, refers to specific hybridization between the ASO or AR nucleotide sequence and a target nucleotide sequence that is complementary within the ranges set forth herein. In some examples, specific hybridization occurs where, under ex vivo conditions, the hybridization occurs under high stringency conditions. By “high stringency conditions” is meant that the ASO or AR, under such ex vivo conditions, hybridize to a target sequence in an amount that is detectably stronger than non-specific hybridization. High stringency conditions, then, are conditions that distinguish a polynucleotide with an exact complementary sequence, or one containing only a few scattered mismatches from a random sequence that happened to have a few small regions (e.g., 1-5 bases) that matched the probe. Such small regions of complementarity are more easily melted than a full-length complement of 12-17 or more bases, and moderate stringency hybridization makes them easily distinguishable. In one example, high stringency conditions include, for example, low salt and/or high temperature conditions, such as provided by about 0.02-0.1 M NaCl or the equivalent, at temperatures of about 50-70° C. The skilled person will appreciate that under in vivo conditions, the specificity of hybridization between an ASO or an AR and its target sequence is defined in terms of the level of complementarity between the ASO or an AR and the target sequence to which it hybridizes within a cell.

[0051] The term “peptide” is intended to include compounds composed of amino acid residues linked by amide bonds. A peptide may be natural or unnatural, ribosome encoded or synthetically derived. Typically, a peptide will consist of between 2 and 200 amino acids. For example, the peptide may have a length in the range of 10 to 20 amino acids or 10 to 30 amino acids or 10 to 40 amino acids or 10 to 50 amino acids or 10 to 60 amino acids or 10 to 70 amino acids or 10 to 80 amino acids or 10 to 90 amino acids or 10 to 100 amino acids, including any length within said range(s). The peptide may comprise or consist of fewer than about 150 amino acids or fewer than about 125 amino acids or fewer than about 100 amino acids or fewer than about 90 amino acids or fewer than about 80 amino acids or fewer than about 70 amino acids or fewer than about 60 amino acids or fewer than about 50 amino acids.

[0052] Peptides, as referred to herein, include “inverso” peptides in which all L-amino acids are substituted with the corresponding D-amino acids, “retro-inverso” peptides in which the sequence of amino acids is reversed and all L-amino acids are replaced with D-amino acids.

[0053] Peptides may comprise amino acids in both L- and/or D-form. For example, both L- and D-forms may be used for different amino acids within the same peptide sequence. In some examples the amino acids within the peptide sequence are in L-form, such as natural amino acids. In some examples the amino acids within the peptide sequence are a combination of L- and D-form.

Further, peptides may comprise unusual, but naturally occurring, amino acids including, but not limited to, hydroxyproline (Hyp), beta-alanine, citrulline (Cit), ornithine (Orn), norleucine (Nle), 3-nitrotyrosine, nitroarginine, pyroglutamic acid (Pyr). Peptides may also incorporate unnatural amino acids including, but not limited to, homo amino acids, N-methyl amino acids, alpha-methyl amino acids, beta (homo) amino acids, gamma amino acids, and N-substituted glycine. Peptides may be linear peptides or cyclic peptides.

[0054] The term “protein” shall be taken to include a single polypeptide chain, i.e., a series of contiguous amino acids linked by peptide bonds or a series of polypeptide chains covalently or non-covalently linked to one another (i.e., a polypeptide complex). For example, the series of polypeptide chains can be covalently linked using a suitable chemical bond or a disulfide bond. Examples of non-covalent bonds include hydrogen bonds, ionic bonds, Van der Waals forces, and hydrophobic interactions.

[0055] Percentage amino acid sequence identity with respect to a given amino acid sequence is defined as the percentage of amino acid residues in a candidate sequence that are identical to the amino acid residues in the reference sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity. Amino acid sequence identity may be determined using the EMBOSS Pairwise Alignment Algorithms tool available from The European Bioinformatics Institute (EMBL-EBI), which is part of the European Molecular Biology Laboratory. This tool is accessible at the website located at [www.ebi.ac.uk/Tools/emboss/align/](http://www.ebi.ac.uk/Tools/emboss/align/). This tool utilizes the Needleman-Wunsch global alignment algorithm (Needleman and Wunsch, 1970). Default settings are utilized which include Gap Open: 10.0 and Gap Extend 0.5. The default matrix “Blosom62” is utilized for amino acid sequences and the default matrix.

[0056] The term “cell penetrating peptide” (CPP) refers to a peptide that is capable of crossing a cellular membrane. In one example, a CPP is capable of translocating across a mammalian cell membrane and entering into a cell. In another example, a CPP may direct a conjugate to a desired subcellular compartment. Thus, a CPP may direct or facilitate penetration of a molecule of interest across a phospholipid, mitochondrial, endosomal, lysosomal, vesicular, or nuclear membrane. A CPP may be translocated across the membrane with its amino acid sequence complete and intact, or alternatively partially degraded.

[0057] A CPP may direct a molecule of interest, such as an ASO disclosed herein, from outside a cell through the plasma membrane, and into the cytoplasm or a desired subcellular compartment. Alternatively, or in addition, a CPP may direct a molecule of interest across the, epithelial,



endothelial, basement membrane, trans-mucosal, cardiovascular, skin, gastrointestinal and/or pulmonary barriers. In some embodiments the CPP is selectively targeted to or taken up by the kidneys.

[0058] The term “peptide ligand” or “receptor binding domain” refers to a peptide that is capable of binding to a membrane surface receptor to enable translocation of the peptide across a cellular membrane. In one example a peptide ligand may enable translocation across the cellular membrane via the natural endocytosis of the targeted receptor. In another example the peptide ligand may utilise a complementary mechanism of translocation across the cellular membrane including utilising a conjugated CPP. In one example, a peptide ligand is capable of translocating across a mammalian cell membrane and to enter a cell. In another example, a peptide ligand may direct a conjugate to a desired subcellular compartment. Thus, a peptide ligand may direct or facilitate cellular uptake of a molecule of interest across a phospholipid, mitochondrial, endosomal, lysosomal, vesicular, or nuclear membrane. A peptide ligand may be translocated across the membrane with its amino acid sequence complete and intact, or alternatively partially degraded.

[0059] A peptide ligand via its binding to a target receptor may direct a molecule of interest, such as an ASO disclosed herein, from outside a cell through the plasma membrane, and into the cytoplasm or a desired subcellular compartment. Alternatively, or in addition, a peptide ligand via its binding to a target receptor may direct a molecule of interest across a relevant biological barrier, e.g., the renal basement membrane, blood-brain, trans-mucosal, hematoretinal, cardiovascular, skin, gastrointestinal, and/or pulmonary barriers.

#### Compositions for Increasing PKD1 mRNA and Polycystin-1 Protein Levels

[0060] MicroRNAs (miRNAs) are a family of short (19 to 23 nucleotide) non-coding single stranded labile RNAs. Although they can bind to any part of target mRNA, their main mode of action is to bind to complementary RNA sequences in the 3' untranslated region (3' UTR) and regulate gene expression by stimulating either mRNA degradation or translational repression. Both mechanisms lead to diminished expression of the target gene. In the case of the Polycystic Kidney Disease 1 (PKD1) gene that encodes Polycystin 1 protein, miRNAs that bind to the 3' UTR and of the encoding mRNA transcript include miR-17 family miRNAs (e.g., miR-17-5p, miR-106a-5p, miR-106b-5p, miR-20a-5p, miR-93-5p) and miR-200 family miRNAs (e.g., miR-200b, miR200c, or miR-429).

[0061] While not wishing to be bound by theory, it is believed that an antisense sequence at least partly complementary to the binding sites for microRNAs located within the PKD1 mRNA 3' UTR will hybridize to the PKD1 3' UTR and sterically hinder (“mask”) access of these miRNAs to their binding sites on the PKD1 3' UTR, ultimately allowing increases in Polycystin 1 protein levels. Accordingly, disclosed herein is an ASO that binds to a targeted portion of the 3' UTR of a PKD1 mRNA wherein binding of the antisense oligonucleotide to the targeted portion increases the level of PKD1 mRNA and/or Polycystin 1 protein. In some preferred examples the ASO hybridizes to a targeted portion of the 3' UTR of PKD1 mRNA, whereby one or more miRNAs are unable to hybridize to their specific target sequences and consequently, increased levels of PKD1 mRNA and Polycystin 1 protein.

[0062] Also disclosed herein is a vector for expression, in a mammalian cell, of an AR that binds to a targeted portion of the 3' UTR of PKD1 mRNA ultimately resulting in increased levels of Polycystin 1 protein as described above.

[0063] For reference, the sequence of the canonical human PKD1 mRNA transcript (“PKD-201”) is publicly available through the online Ensembl database under record ENST00000262304.9. The nucleotide sequence of the canonical human PKD1 mRNA 3' UTR sequence is provided herein as SEQ ID NO:1.

#### Antisense Oligonucleotides (ASOs) and Antisense RNAs (ARs)

[0064] In some preferred examples of the compositions and methods described herein, ASOs and ARs have a sequence that is completely complementary across its length to the target sequence or a

sequence near complementarity (e.g., sufficient complementarity to bind the target sequence and interfere with miRNA binding at the PKD1 mRNA 3' UTR binding site). ASOs and ARs are designed so that they bind (hybridize) to a target RNA sequence (e.g., a targeted portion of a pre-mRNA transcript) and remain hybridized under physiological conditions. Selection of suitable sequences for ASOs and ARs generally avoids, where possible, similar nucleic acid sequences in other (i.e., off-target) locations in the genome or in cellular mRNAs or miRNAs, such that the likelihood the ASO or AR will hybridize at such sites is limited.

[0065] In some examples, ASOs or ARs “specifically hybridize” to or are “specific” to a target nucleic acid or a targeted portion of the PKD1 mRNA 3' UTR. At a given ionic strength and pH, the T<sub>sub</sub>m is the temperature at which 50% of a target sequence hybridizes to a complementary oligonucleotide.

[0066] ASO and AR sequences are “complementary” to their target sequences when hybridization occurs in an antiparallel configuration between two single-stranded polynucleotides of complementary sequence. Complementarity is quantifiable in terms of the proportion (e.g., the percentage) of bases in opposing strands that are expected to form hydrogen bonds with each other, according to generally accepted base-pairing rules. The nucleotide sequence of an ASO or AR need not be 100% complementary to that of its target nucleic acid to hybridize. In certain examples, the nucleotide sequences of ASOs or ARs in the compositions disclosed herein can be at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% sequence complementary to the nucleotide sequence of the targeted portion of an RNA transcript over the length of the ASO or AR nucleotide sequence. For example, an ASO or AR in which 18 of 20 nucleotides of ASO or AR sequence are complementary to a target region, and would therefore specifically hybridize, would represent 90 percent complementarity. In such an example, the remaining non-complementary nucleotides of the ASO or AR could be clustered together or interspersed with complementary nucleotides and need not be contiguous. Complementarity of an ASO or AR sequence to a target nucleotide sequence (expressed as “percent complementarity” to its target sequence; or “percent identity” to its reverse complement sequence) can be determined routinely using algorithms known in the art, as exemplified in the BLAST programs (basic local alignment search tools) and PowerBLAST programs (Altschul, et al., 1990, *J. Mol. Biol.*, 215:403-410; Zhang et al., 1997, *Genome Res.*, 7:649-656).

[0067] In some examples, an ASO or AR does not hybridize to all nucleotides in a target sequence and the nucleotide positions at which it does hybridize may be contiguous or non-contiguous. ASOs or ARs may hybridize over one or more segments of a 3' UTR region of a mRNA, such that intervening or adjacent segments are not involved in the hybridization event (e.g., a loop structure or hairpin structure may be formed).

[0068] In some examples the nucleotide sequences of ASOs or ARs described herein are complementary to a targeted portion of PKD1 mRNA 3' UTR. In some preferred examples, the ASOs or ARs are complementary to a targeted portion of the PKD1 mRNA 3' UTR corresponding to SEQ ID NO:1. In some examples the nucleotide sequence of the ASO or the AR is at least 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 100% complementary to the nucleotide sequence of the targeted portion of the PKD1 3' UTR over the length of the ASO or the AR. In some examples the nucleotide sequence of the ASO or the AR is at least 60%, 61%, 62%, 63%, 64%, 65%, 66%, 67%, 68%, 69%, 70%, 71%, 72%, 73%, 74%, 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or 100% sequence identity, over the entire length of any of the sequences provided in SEQ ID NOs: 2-351 or 353-362 (shown in Table 1).

[0069] The ASOs or ARs for use in the compositions described herein may be of any length suitable for specific hybridization to a target sequence. In some examples, the nucleotide sequence of the ASOs or ARs consist of 8 to 50 nucleotides. For example, the ASO or AR sequence can be

10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 40, 45, or 50 nucleotides in length. In some examples, the ASOs consist of more than 50 nucleotides, but no more than 100 nucleotides in length. In some examples, the ASO or AR nucleotide sequence is from 8 to 50 nucleotides, 8 to 40 nucleotides, 8 to 35 nucleotides, 8 to 30 nucleotides, 8 to 25 nucleotides, 8 to 20 nucleotides, 8 to 15 nucleotides, 9 to 50 nucleotides, 9 to 40 nucleotides, 9 to 35 nucleotides, 9 to 30 nucleotides, 9 to 25 nucleotides, 9 to 20 nucleotides, 9 to 15 nucleotides, 10 to 50 nucleotides, 10 to 40 nucleotides, 10 to 35 nucleotides, 10 to 30 nucleotides, 10 to 25 nucleotides, 10 to 20 nucleotides, 10 to 15 nucleotides, 11 to 50 nucleotides, 11 to 40 nucleotides, 11 to 35 nucleotides, 11 to 30 nucleotides, 11 to 25 nucleotides, 11 to 20 nucleotides, 11 to 15 nucleotides, 12 to 50 nucleotides, 12 to 40 nucleotides, 12 to 35 nucleotides, 12 to 30 nucleotides, 12 to 25 nucleotides, 12 to 20 nucleotides, 12 to 15 nucleotides, 13 to 50 nucleotides, 13 to 40 nucleotides, 13 to 35 nucleotides, 13 to 30 nucleotides, 13 to 25 nucleotides, 13 to 20 nucleotides, 14 to 50 nucleotides, 14 to 40 nucleotides, 14 to 35 nucleotides, 14 to 30 nucleotides, 14 to 25 nucleotides, 14 to 20 nucleotides, 15 to 50 nucleotides, 15 to 40 nucleotides, 15 to 35 nucleotides, 15 to 30 nucleotides, 15 to 25 nucleotides, 15 to 20 nucleotides, 20 to 50 nucleotides, 20 to 40 nucleotides, 20 to 35 nucleotides, 20 to 30 nucleotides, 20 to 25 nucleotides, 25 to 50 nucleotides, 25 to 40 nucleotides, 25 to 35 nucleotides, or 25 to 30 nucleotides in length. In some examples, the ASOs or ARs are 20 nucleotides in length. In some preferred examples, the nucleotide sequence of the ASO or AR nucleotide is 25 nucleotides in length.

[0070] In some examples for each occurrence of “G” in an ASO or AR sequence disclosed herein, the “G” is guanosine or inosine. In some examples for each occurrence of “T” in an ASO or AR sequence disclosed herein, the “T” is any one of: thymidine, inosine, uracil, or an isomeric or modified form of uracil (e.g., pseudouridine or N1-methyl-pseudouridine). In some examples for each occurrence of “C” in an ASO or AR sequence disclosed herein, the C is cytosine or a modified form of cytosine (e.g., 5'-methyl cytosine).

[0071] In some examples the nucleotide sequence of the ASO or AR comprises the sequence of any one of SEQ ID NOs:2-351 or 353-362. In some examples the nucleotide sequence of the ASO or AR consists of the sequence of any one of SEQ ID NOs:2-351 or 353-362. In some examples the nucleotide sequence of the ASO or AR comprises or consists of the sequence of any one of SEQ ID NOs:17, 47, 73, 334-337, or 355-360. In some examples the nucleotide sequence of the ASO or AR comprises or consists of the sequence of any one of SEQ ID NOs:334 or 335. In some examples the nucleotide sequence of the ASO or AR comprises or consists of the sequence of any one of SEQ ID NOs:355-360. In some examples the nucleotide sequence of the ASO or AR comprises or consists of the sequence of any one of SEQ ID NOs:17, 73, 336, and 337. In some examples the nucleotide sequence of the ASO or AR comprises or consists of SEQ ID NO:47.

#### ASO Chemistry and Modifications

[0072] The ASOs used in the compositions described herein may comprise naturally-occurring nucleotides, nucleotide analogues, modified nucleotides, or any combination thereof. The term “naturally occurring nucleotides” includes deoxyribonucleotides and ribonucleotides. The term “modified nucleotides” includes nucleotides with modified or substituted sugar groups and/or having a modified backbone. In some examples, all the nucleotides of an ASO are modified nucleotides. Chemical modifications of ASOs or components of ASOs that are compatible with the compositions and methods described herein are known in the art as disclosed in, e.g., in U.S. Pat. Nos. 8,258,109, 5,656,612, U.S. Patent Publication No. 2012/0190728, and Roberts et al., 2020, Nature Rev. Drug Disc., 19:673-694.

[0073] One or more nucleotides of an ASO may be any naturally occurring, unmodified nucleobase such as adenine, guanine, cytosine, thymine, uracil and inosine, or any synthetic or modified nucleobase that is sufficiently similar to an unmodified nucleobase such that it is capable of hydrogen bonding with a nucleobase present on a target RNA transcript. Examples of suitable modified nucleobases include, but are not limited to, hypoxanthine, xanthine, 7-methylguanine, 5,

6-dihydrouracil, 5-methylcytosine, and 5-hydroxymethylcytosine.

[0074] ASOs include a “backbone” structure, that refers to the connection between nucleotides/monomers of the ASO. In naturally occurring oligonucleotides, the backbone comprises a 3'-5' phosphodiester linkage connecting sugar moieties of adjacent nucleotides. Suitable types of backbone linkages for the ASOs described herein include, but are not limited to, phosphodiester, phosphorothioate, phosphorodithioate, phosphorodiamidate, phosphoroselenoate, phosphorodiselenoate, phosphoroanilothioate, phosphoraniladate, phosphoramidate, and the like. In some examples, the backbone modification is a phosphorothioate linkage. In other examples, the backbone modification is a phosphorodiamidate linkage. See, e.g., Roberts et al. *supra*; and Agrawal (2021), *Biomedicines*, 9:503. In some examples, the backbone structure of the ASO does not contain phosphorous-based linkages, but rather contains peptide bonds, for example in a peptide nucleic acid (PNA), or linking groups including carbamate, amides, and linear and cyclic hydrocarbon groups.

[0075] In some examples, the stereochemistry at each of the phosphorus internucleotide linkages of the ASO backbone is random. In other examples, the stereochemistry at each of the phosphorus internucleotide linkages of the ASO backbone is controlled and is not random. For example, U.S. Pat. No. 9,605,019 describes methods for independently selecting the handedness of chirality at each phosphorous atom in an oligonucleotide. In some examples, an ASO used in the compositions and methods provided herein, including, but not limited to, the ASOs the sequences of which are disclosed herein as SEQ ID NOs:2-351 is an ASO having phosphodiester internucleotide linkages that are not random. In some examples, a composition or composition used in the methods disclosed herein comprises a pure diastereomeric ASO. In other examples, the composition comprises an ASO that has diastereomeric purity of at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, at least about 99%, about 100%, about 90% to about 100%, about 91% to about 100%, about 92% to about 100%, about 93% to about 100%, about 94% to about 100%, about 95% to about 100%, about 96% to about 100%, about 97% to about 100%, about 98% to about 100%, or about 99% to about 100%.

[0076] In some examples, the ASO has a non-random mixture of Rp and Sp configurations at its phosphorus internucleotide linkages. In some examples, an ASO used in the compositions and methods disclosed herein, comprises about 5-100% Rp, at least about 5% Rp, at least about 10% Rp, at least about 15% Rp, at least about 20% Rp, at least about 25% Rp, at least about 30% Rp, at least about 35% Rp, at least about 40% Rp, at least about 45% Rp, at least about 50% Rp, at least about 55% Rp, at least about 60% Rp, at least about 65% Rp, at least about 70% Rp, at least about 75% Rp, at least about 80% Rp, at least about 85% Rp, at least about 90% Rp, or at least about 95% Rp, with the remainder Sp, or about 100% Rp.

[0077] In some examples, the ASOs described herein contain a sugar moiety that comprises ribose or deoxyribose, or a modified sugar moiety or sugar analogue, including a morpholine ring. Suitable examples of modified sugar moieties include, but are not limited to, 2' substitutions such as 2'-O-modifications, 2'-O-methyl (2'-O-Me), 2'-O-methoxyethyl (2'MOE), 2'-O-aminoethyl, 2'F, N3'->P5' phosphoramidate, 2'dimethylaminooxyethoxy, 2'dimethylaminoethoxyethoxy, 2'-guanidinidium, 2'-O-guanidinium ethyl, carbamate modified sugars, and bicyclic modified sugars. In some examples, the sugar moiety modification is selected from among 2'-O-Me, 2'F, and 2'MOE. In other examples, the sugar moiety modification is an extra bridge bond, such as in a locked nucleic acid (LNA). In some examples the sugar analogue contains a morpholine ring, such as phosphorodiamidate morpholino (PMO). In some examples, the sugar moiety comprises a ribofuransyl or 2'deoxyribofuransyl modification. In some examples, the sugar moiety comprises 2'4'-constrained 2'-O-methoxyethyl (cMOE) modifications. In some examples, the sugar moiety comprises cEt 2', 4' constrained 2'-O ethyl BNA modifications. In other examples, the sugar moiety comprises tricycloDNA (tcDNA) modifications. In some examples, the sugar moiety comprises

ethylene nucleic acid (ENA) modifications. In some examples, the sugar moiety comprises 2'-O-(2-N-methylcarbamoyl) (MCE). Modifications are known in the art as exemplified in Jarver, et al., 2014, *Nucleic Acid Therapeutics*, 24(1): 37-47.

[0078] In some examples, each constituent nucleotide of the ASO is modified in the same way, e.g., every linkage of the backbone of the ASO comprises a phosphorothioate linkage, or each ribose sugar moiety comprises a 2'-O-methyl modification. In other examples, a combination of different modifications is used, e.g., an ASO comprising a combination of phosphorodiamidate linkages and sugar moieties comprising morpholine rings (morpholinos).

[0079] In some examples, the ASO comprises one or more backbone modifications. In some examples, the ASO comprises one or more sugar moiety modification. In some examples, the ASO comprises one or more backbone modifications and one or more sugar moiety modifications. In some examples, the ASO comprises a 2'-O-MOE modification and a phosphorothioate backbone. In some examples, the ASO comprises a peptide nucleic acid (PNA).

[0080] In some preferred examples, the ASO comprises a phosphorodiamidate morpholino (PMO).

[0081] The skilled person in the art will appreciate that ASOs may be modified, in order to achieve desired properties or activities of the ASO or reduce undesired properties or activities of the ASO. In some examples, an ASO is modified to alter one or more properties. For example, such modifications can: enhance binding affinity to a target sequence on a pre-mRNA transcript; reduce binding to any non-target sequence; reduce degradation by cellular nucleases (e.g., RNase H); improve uptake of an ASO into a cell and/or particular subcellular compartments; alter the pharmacokinetics or pharmacodynamics of the ASO; and/or modulate the half-life of the ASO in vivo.

[0082] In some examples, the ASOs comprise one or more 2'-O-(2-methoxyethyl) (MOE) phosphorothioate-modified nucleotides that have been shown to confer significantly enhanced resistance of ASOs to nuclease degradation and increased bioavailability.

[0083] Methods for synthesis and chemical modification of ASOs, as well as synthesis of ASO conjugates is well known in the art, and such ASOs are available commercially.

[0084] In some examples, a composition (e.g., a pharmaceutical composition) provided here includes two or more ASOs with different chemistries but complementary to the same targeted portion of the PKD1 mRNA 3' UTR. In other examples, two or more ASOs that are complementary to different targeted portions of the PKD1 mRNA 3' UTR.

[0085] In some examples, the compositions disclosed herein include ASOs that are linked to a functional moiety. In some examples, the functional moiety is a delivery moiety, a targeting moiety, a detection moiety, a stabilizing moiety, or a therapeutic moiety. In some examples the functional moiety includes a delivery moiety or a targeting moiety. In some examples the functional moiety includes a stabilizing moiety. In some preferred examples the functional moiety is a delivery moiety.

[0086] Suitable delivery moieties include, but are not limited to, lipids, peptides, carbohydrates, polyethers, and antibodies.

[0087] In some examples, the delivery moiety includes a cell-penetrating peptide (CPP).

[0088] Suitable examples of CPPs are described in, e.g., PCT/AU2020/051397. In some examples the amino acid sequence of the CPP comprises or consists of: RRSRTARAGRPGRNSSRPSAPR (SEQ ID NO: 352). In one example, the CPP comprises the sequence RRSRTARAGRPGRNSSRPSAPR (SEQ ID NO: 352), optionally wherein any amino acid other than glycine is a D amino acid. In other examples, the delivery moiety includes a receptor binding domain.

[0089] In other examples, the delivery moiety includes a carbohydrate. In some examples, a carbohydrate delivery moiety is selected from among N-acetylgalactosamine (GalNAc), N-Ac-Glucosamine (GluNAc), and a mannose. In one example, the carbohydrate delivery moiety is GalNAc.

[0090] In other examples, the delivery moiety includes a lipid. Examples of suitable lipids as delivery moieties include, but are not limited to, cholesterol moiety, a cholesteryl moiety, and aliphatic lipids. In some examples the delivery moiety includes a fatty acid or lipid moiety. In some embodiments the fatty acid chain length is about C8 to C20. Examples of suitable fatty acid moieties and their conjugation to oligonucleotides are found in, e.g., International Patent Publication WO 2019232255 and in Prakash et al., (2019).

[0091] In further examples, the delivery moiety includes an antibody, as described in, e.g., Dugal-Tessier et al., (2021).

[0092] Suitable examples of stabilizing moieties include, but are not limited to, polyethylene glycol (PEG), poly(oligo(ethylene glycol) methyl ether methacrylate) (POEGMA), and Poly(2-oxazoline)s (POx).

[0093] In some examples, where an ASO is linked to a functional moiety, the functional moiety is covalently linked to the ASO. In other examples, the functional moiety is non-covalently linked to the ASO.

[0094] Functional moieties can be linked to one or more of any nucleotides in an ASO at any of several positions on the sugar, base or phosphate group, as understood in the art and described in the literature, e.g., using a linker. Linkers can include a bivalent or trivalent branched linker. In some examples, the functional moiety is linked to the 5' end of the ASO. In other examples, the functional moiety is linked to the 3' end of the ASO.

[0095] In some examples compositions comprising any of the ASOs disclosed herein also include a delivery nanocarrier complexed with ASO. In some examples, a delivery nanocarrier is selected from among lipoplexes, liposomes, exosomes, inorganic nanoparticles, and DNA nanostructures. In other examples the delivery nanocarrier includes a lipid nanoparticle encapsulating the ASO.

Various delivery ASO-nanocarrier complex formats are known in the art, as reviewed in, e.g., Roberts et al., supra.

#### Vectors for Expression of PKD1 Antisense RNA (AR)

[0096] In some examples provided herein are compositions containing a vector for expression, in a mammalian cell, of an AR that binds to a targeted portion of the PKD1 3' UTR, wherein binding of the AR to the targeted portion reduces specific binding of miRNA binding to the 3' UTR, e.g., binding of miR-17 family miRNAs (e.g., miR-17-5p, miR-106a-5p, miR-106b-5p, miR-20a-5p, miR-93-5p) or miR-200 family miRNAs (e.g., miR-200b, miR200c, or miR-429).

[0097] In some examples, the promoter used in the expression vector is a kidney cell type-selective promoter for driving expression of the AR in the mammalian cell. In some examples, the kidney cell type-selective promoter is selective for expression in a kidney cell type selected from the list consisting of: pericytes, podocytes, parietal epithelial cells, proximal tubule cells, ascending loop of Henle cells, descending loop of Henle cells, distal tubule cells, connecting tubule cells, intercalated cells, principal cells, peritubular capillary endothelium cells, and glomerular endothelium cells. For example, the sodium-dependent phosphate transporter type 2a (NPT2a) promoter for expression in the proximal tubule, the sodium-potassium-2-chloride cotransporter (NKCC2) promoter for expression in the thick ascending limb of Henle (TALH), the aquaporin 2 (AQP2) promoter for expression in the collecting duct, and the podocin promoter for expression in podocytes.

[0098] In some examples the promoter is an inducible promoter, e.g., inducible by a ligand-regulated transactivator such as the Tet-inducible rtTA that allows titration of AR transcription in a target mammalian cell. In some examples, the promoter driving AR expression is a U6 or other Pol III promoter, which is particularly suitable for transcription of short RNA sequences such as AR sequences disclosed herein. In some examples, an expression vector utilizes hybrid promoter systems, e.g., a Tet-O-regulated U6 promoter system as described in Lin et al. (2004), FEBS Letters, 577 (2004) 376-380. In some examples, where both cell type-specificity and inducibility of an AR expression vector are desired, a two-part expression system is used in which expression of a ligand-regulated transactivator is driven by a cell type-selective promoter and expression of an AR

disclosed herein is driven by a promoter regulated by the ligand-regulated transactivator.

[0099] In some examples, the expression vectors used in the compositions disclosed herein are non-viral expression vectors, e.g., plasmid vectors, minicircle DNA vectors, linear amplicon expression cassettes, and the like.

[0100] In some examples, composition containing a non-viral expression virus further comprises a transfection agent. Exemplary transfection agents for transfection include, but are not limited to, jet-PEI® (available from Polyplus-Transfection® SA, Strasbourg, France); TurboFect in vivo Transfection Reagent (ThermoFisher), and cationic derivatives of polyisoprenoid alcohols (PTAI) as described in, e.g., Rak et al. (2016), *J Gene Med*, 18(11-12):331-342.

[0101] In other examples, the expression vectors to be used are viral vectors, i.e., non-replicative recombinant viruses suitable for expression of an AR disclosed herein.

[0102] Preferably, the recombinant virus for expression of the PKD1 AR is a DNA virus. Suitable types of DNA viruses include adeno-associated virus (AAV), adenovirus, lentivirus, herpes simplex virus (HSV), and anelloviruses. Methods for design, production, and use of such types of recombinant DNA viruses are established in the art, as exemplified in Fukazawa et al., (2010), *International J of Mol. Med*, 25(1), 3-10, and in “Gene Therapy Protocols” for adenovirus; “Adeno-Associated Virus: Methods and Protocols” for AAV; Cody et al (2013), *Journal of Genetic Syndromes & Gene Therapy*, 4(1), 126, and “Herpes Simplex Virus: Methods and Protocols” for HSV; “Gene Therapy Protocols Vol. 1: Production and In Vivo Applications of Gene Transfer Vectors”; and Merten et al. (2016), *Molecular Therapy—Methods & Clinical Development*, 3, 16017, and Emeagi et al. (2013), *Current Molecular Medicine* 13(4), 602-625 for lentivirus. In some preferred examples, the viral vector is a recombinant AAV.

#### Genetically Modified Cells

[0103] Also provided herein are genetically modified cells. In some examples the genetically modified cells are genetically modified bacterial cells (e.g., recombinant *E. coli*, for amplifying an AR expression vector disclosed herein). In other examples the genetically modified cells are genetically modified mammalian cells that have been transfected with any of the ASOs or non-viral AR expression vectors; or transduced with any of the viral AR expression vectors disclosed herein. In some examples, the genetically modified mammalian cells are ex vivo, e.g., as a cultured cell population. In other examples, the genetically modified mammalian cells are in vivo, e.g., in a mouse. In some examples, the genetically modified mammalian cells are human cells.

[0104] In some examples the genetically modified mammalian cells are of primary cell types. Suitable examples of primary cell types include, but are not limited to, cyst cells, pericytes, podocytes, parietal epithelial cells, proximal tubule cells, ascending loop of Henle cells, descending loop of Henle cells, distal tubule cells, connecting tubule cells, intercalated cells, principal cells, peritubular capillary endothelium cells, and glomerular endothelium cells. In some examples such primary cell types can be obtained by differentiation of a human pluripotent stem cell line, e.g., a human induced pluripotent stem cell (hiPSC) line or a human embryonic stem cell (hESC) line. Methods for obtaining a variety of different renal cell types is known in the art, as reviewed in, e.g., de Carvalho Ribeiro et al. (2020) and Osafune et al. (2021). In some examples, the primary cell types are derived from dividing kidney tissue such as renal cysts. Cyst cells excised from ADPKD kidneys have been extensively used since the 1980s. A detailed protocol for in vitro cyst formation was recently published by Sharma et al. (2019). In other examples, the genetically modified mammalian cells are derived from a cell line. In some examples the cell line is pluripotent stem cell line (e.g., hiPSCs or hESCs) or a human kidney cell line. In some examples, the genetically modified mammalian cells are derived from HEK293, HK-2 or WT9-7 cell lines. In some preferred examples, the genetically modified mammalian cells express Polycystin 1 endogenously. In some embodiments genetically modified human cells are provided in a kidney organoid, e.g., a kidney organoid derived by differentiation of human pluripotent or human adult stem cells as reviewed in, e.g., Kang (2023), *Development & Reproduction*, 27(2):57-65.

[0105] The genetically modified cells disclosed herein can be genetically modified by any of a number of methods and strategies known in the art, e.g., transient transfection, stable transfection, and viral transduction. In some examples transfection with ASOs or non-viral vectors is carried out by nucleofection. In other examples transfection of cells is by lipofection.

#### Pharmaceutical Compositions

[0106] Also provided herein are pharmaceutical compositions comprising any of the foregoing ASOs, non-viral expression vectors, modified messenger RNAs (mmRNAs), and viral expression vectors disclosed herein, and formulated with at least a pharmaceutically acceptable excipient, including a carrier, filler, preservative, adjuvant, solubilizer and/or diluent.

[0107] Pharmaceutical compositions containing any of the ASOs or expression vector compositions described herein, for use in the methods disclosed herein, can be prepared according to conventional techniques well known in the pharmaceutical industry and described in the published literature. In some examples, a pharmaceutical composition for treating a subject comprises a therapeutically effective amount of any ASO or expression vector disclosed herein.

[0108] Pharmaceutically acceptable salts are suitable for use in contact with the tissues of humans and lower animals without undue toxicity, irritation, allergic response, etc., and are commensurate with a reasonable benefit/risk ratio. Examples of pharmaceutically acceptable, nontoxic acid addition salts are salts of an amino group formed with inorganic acids such as hydrochloric acid, hydrobromic acid, phosphoric acid, sulfuric acid and perchloric acid or with organic acids such as acetic acid, oxalic acid, maleic acid, tartaric acid, citric acid, succinic acid or malonic acid. Other pharmaceutically acceptable salts include adipate, alginate, ascorbate, aspartate, benzenesulfonate, benzoate, bisulfate, borate, butyrate, camphorate, camphorsulfonate, citrate, cyclopentanepropionate, digluconate, dodecylsulfate, ethanesulfonate, formate, fumarate, glucoheptonate, glycerophosphate, gluconate, hemisulfate, heptanoate, hexanoate, hydroiodide, 2-hydroxy-ethanesulfonate, lactobionate, lactate, laurate, lauryl sulfate, malate, maleate, malonate, methanesulfonate, 2-naphthalenesulfonate, nicotinate, nitrate, oleate, oxalate, palmitate, pamoate, pectinate, persulfate, 3-phenylpropionate, phosphate, picrate, pivalate, propionate, stearate, succinate, sulfate, tartrate, thiocyanate, p-toluenesulfonate, undecanoate, valerate salts, and the like. Representative alkali or alkaline earth metal salts include sodium, lithium, potassium, calcium, magnesium, and the like. Further pharmaceutically acceptable salts include, when appropriate, nontoxic ammonium, quaternary ammonium, and amine cations formed using counterions such as halide, hydroxide, carboxylate, sulfate, phosphate, nitrate, lower alkyl sulfonate and aryl sulfonate.

[0109] In some examples, pharmaceutical compositions are formulated into any of a number of possible dosage forms including, but not limited to, topical ointments, solutions for intravenous administration, subcutaneous injection, intrathecal administration, intracisterna magna administration, tablets, capsules, gel capsules, liquid syrups, and soft gels. In some examples, the compositions are formulated as suspensions in aqueous, non-aqueous or mixed media. Aqueous suspensions may further contain substances that increase the viscosity of the suspension including, for example, sodium carboxymethylcellulose, sorbitol and/or dextran. The suspension may also contain stabilizers. In some examples, a pharmaceutical formulation disclosed herein is provided in a form including, but not limited to, a solution, emulsion, microemulsion, foam or liposome-containing formulation (e.g., cationic or noncationic liposomes).

[0110] In some examples, pharmaceutical formulations comprising any of the ASOs or expression vectors described herein may comprise one or more penetration enhancers, carriers, excipients or other active or inactive ingredients as appropriate and known to the skilled person. In some examples, where a pharmaceutical composition includes liposomes, such liposomes can also include sterically stabilized liposomes, e.g., liposomes comprising one or more specialized lipids. These specialized lipids result in liposomes with enhanced circulation lifetimes. In some examples, a sterically stabilized liposome comprises one or more glycolipids or is derivatized with one or more hydrophilic polymers, such as PEG moiety. In some examples, a surfactant is included in the



pharmaceutical formulation.

[0111] In some examples, a pharmaceutical composition also includes a penetration enhancer to enhance the delivery of ASOs or non-viral expression vectors, e.g., to aid diffusion across cell membranes and/or enhance the permeability of a lipophilic drug. In some examples, the penetration enhancers include a surfactant, a fatty acid, a bile salt, or a chelating agent.

[0112] In some examples, a pharmaceutical composition comprises a dose of ASOs or non-viral vectors ranging from about 0.0001 mg/kg to about 80 mg/kg, e.g., 0.005 mg/kg, 0.007 mg/kg, 0.01 mg/kg, 0.02 mg/kg, 0.03 mg/kg, 0.05 mg/kg, 0.1 mg/kg, 0.2 mg/kg, 0.5 mg/kg, 1 mg/kg, 3 mg/kg, 5 mg/kg, 8 mg/kg, 10 mg/kg, 15 mg/kg, 20 mg/kg, 25 mg/kg, 30 mg/kg, 35 mg/kg, 40 mg/kg, 45 mg/kg, 50 mg/kg, 55 mg/kg, 60 mg/kg, 65 mg/kg, 70 mg/kg, 77 mg/kg, or another dose ranging from about 0.01 mg/kg to about 80 mg/kg.

[0113] In some examples, a pharmaceutical composition comprises multiple ASOs or AR expression vectors. In some examples, a pharmaceutical composition comprises, in addition to ASOs or AR expression vectors, another drug or therapeutic agent suitable for treatment of a subject suffering from a condition associated with inflammation.

#### Methods

[0114] As described herein, ADPKD is associated with insufficient levels of functional Polycystin 1. Accordingly, the methods described herein include a method treating ADPKD by administering to the subject a therapeutically effective amount of a pharmaceutical composition comprising any of the ASOs or expression vectors disclosed herein. Likewise, in some examples, any of the ASOs or AR expression vectors disclosed herein are used in the manufacture of a medicament for reducing inflammation.

[0115] Also provided herein is a method for increasing the level of PKD1 mRNA and subsequently Polycystin 1 protein in a cell, ex vivo or in a tissue in vivo, the method comprising contacting the cell with an ASO, AR expression vector, or pharmaceutical composition, as disclosed herein, whereby cellular abnormalities associated with ADPKD, e.g., cyst formation are reduced.

[0116] In some examples, administration to a subject or contact with cells with any of the ASOs, AR expression vectors, or pharmaceutical compositions disclosed herein increases the level of PKD1 mRNA or Polycystin 1 protein about 1.1 to about 3-fold, e.g., 1.3 to about 2.7-fold, about 1.4 to about 2.6-fold, about 1.5 to about 2.5-fold, about 1.6 to about 2.4-fold, about 1.7 to about 2.3-fold, about 1.8 to about 2.2-fold, or another increased level of PKD1 mRNA or Polycystin 1 protein about 1.1 to about 3-fold compared to the level in the tissue without the administration or contact.

[0117] Suitable routes of administration for treatment with the compositions, pharmaceutical compositions, or medicaments disclosed herein include, but are not limited to, intravenous, intra-arterial, subcutaneous, intrathecal, oral and topical.

[0118] In some examples any of the treatment methods disclosed herein can, optionally, include the step of determining a level PKD1 mRNA, Polycystin 1 protein in the subject before and/or following the treatment.

[0119] As the skilled person will understand, the treatment methods disclosed herein include administration of the compositions and pharmaceutical compositions disclosed herein in a therapeutically effective amount to a subject (e.g., a human subject). The terms “effective amount” or “therapeutically effective amount,” as used herein, refer to a sufficient amount of a disclosed ASO, non-viral or viral expression vector being administered to relieve to some extent one or more of the symptoms and/or clinical indicia associated with pathological inflammation in a particular disease or health condition. In some examples, an “effective amount” for therapeutic uses is the amount of one of the foregoing agents required to provide a clinically significant decrease in disease symptoms and/or inflammatory markers or to prevent disease symptoms without undue adverse side effects. An appropriate “effective amount” in any individual case may be determined using techniques, such as a dose escalation study. The term “therapeutically effective amount”

includes, for example, a prophylactically effective amount. It is understood that “an effective amount” or “a therapeutically effective amount” can vary from subject to subject, due to variation in metabolism of the compound of any age, weight, general condition of the subject, the condition being treated, the severity of the condition being treated, and the judgment of the prescribing physician. By way of example only, therapeutically effective amounts may be determined by routine experimentation, including but not limited to a dose escalation clinical trial. Where more than one therapeutic agent is used in combination, a “therapeutically effective amount” of each therapeutic agent can refer to an amount of the therapeutic agent that would be therapeutically effective when used on its own or may refer to a reduced amount that is therapeutically effective by virtue of its combination with one or more additional therapeutic agents.

#### Combination Treatments

[0120] The pharmaceutical compositions comprising any of the ASOs or AR expression vectors, disclosed herein, can also be used in combination with other agents of therapeutic value in the treatment of a condition associated with pathological inflammation. In general, other agents do not necessarily have to be administered in the same pharmaceutical composition, and may, because of different physical and chemical characteristics, preferably be administered by different routes. The determination of the mode of administration and the advisability of administration, where possible, in the same pharmaceutical composition, is well within the knowledge of the skilled clinician. The initial administration can be made according to established protocols known in the art, and then, based upon the observed effects, the dosage, modes of administration and times of administration can be modified by the skilled clinician.

[0121] Compositions and pharmaceutical compositions comprising ASOs and/or expression vectors, and an additional therapeutic agent may be administered concurrently (e.g., simultaneously, essentially simultaneously or within the same treatment protocol) or sequentially, depending upon the stage and progression of the inflammatory disease to be treated, the condition of the patient, and the choice of specific therapeutic agents used. The determination of the order of administration, and the number of repetitions of administration of each therapeutic agent during a treatment protocol, is well within the knowledge of the skilled physician after evaluation of the disease being treated and the condition of the patient.

[0122] It is known to those of skill in the art that therapeutically-effective dosages can vary when the drugs are used in treatment combinations. Methods for experimentally determining therapeutically-effective dosages of drugs and other agents for use in combination treatment regimens are described in the literature. For example, the use of metronomic dosing, i.e., providing more frequent, lower doses in order to minimize toxic side effects, has been described extensively in the literature. Combination treatment further includes periodic treatments that start and stop at various times to assist with the clinical management of the patient.

[0123] For combination therapies, dosages of co-administered therapeutic agents will of course vary depending on the type of co-agents employed, ASO or expression vector, and the disease stage of the patient to be treated.

[0124] Pharmaceutical compositions comprising ASOs, AR or expression vectors, and an additional therapeutic agent that make up a combination therapy disclosed herein may be a combined dosage form or in separate dosage forms intended for substantially simultaneous administration. The pharmaceutical compositions that make up the combination therapy may also be administered sequentially, with either therapeutic agent being administered by a regimen calling for two-step administration. The two-step administration regimen may call for sequential administration of the active agents or spaced-apart administration of the separate active agents. The time period between the multiple administration steps may range from, a few minutes to several hours, depending upon the properties of each pharmaceutical agent, such as potency, solubility, bioavailability, plasma half-life and kinetic profile of the pharmaceutical agent. Circadian variation of various physiological parameters may also be evaluated to determine the optimal dose interval.

[0125] Examples of suitable therapeutic agents for co-administration with a composition or a pharmaceutical composition disclosed herein include, but are not limited to, vasopressin V2 receptor antagonists (e.g., Tolvaptan), Angiotensin-converting enzyme (ACE) inhibitors, angiotensin-2 receptor blockers, paracetamol, opioids and antibiotics.

## EXAMPLES

### Example 1: Identification and Sequence Selection of PKD1 3' UTR Target Sequence

[0126] Mutations in PKD1 lead to dysregulation and/or insufficient levels of functional Polycystin 1 associated with ADPKD onset and severity. miRNAs from the miR-17 family (e.g., miR-17-5p, miR-106a-5p, miR-106b-5p, miR-20a-5p, miR-93-5p) and the miR-200 family (e.g., miR-200b, miR200c, or miR-429) are known to bind to the 3' UTR of the PKD1 mRNA leading to an increase in Polycystin 1 protein.

[0127] For a 3' UTR upregulation strategy, PMOs (SEQ ID NOs:2-351 or 353-362) are designed as part of a microwalk strategy to span the entire PKD1 3' UTR (SEQ ID NO:1) encompassing binding sites for miR-17 family (e.g., miR-17-5p, miR-106a-5p, miR-106b-5p, miR-20a-5p, miR-93-5p) and miR-200 family (e.g., miR-200b, miR200c, or miR-429) miRNAs. PMOs having sequences corresponding to SEQ ID NOs:47, 334, and 335 are complementary to the binding site for miR-17 family miRNAs (FIG. 1B), and PMOs having sequences corresponding to SEQ ID NOs:14, 73, 75, 336 and 337 are complementary to the binding sites for miR-200 family miRNAs (FIG. 1C). The miRNA binding sites are highlighted in bold (FIGS. 1B and 1C).

### Example 2: Screening of PPMOs that Target the miR-17 Binding Site in PKD1 in HEK293 Cells for the Ability to Increase PKD1 Transcript Expression

[0128] PMOs with sequences corresponding to PKD1 H46 3UTR(+604+628) (SEQ ID NO:47), PKD1 H46A(+611+635) (SEQ ID NO:334), and PKD1 H46A(+615+639) (SEQ ID NO:335), were conjugated to a cell penetrating peptide (SEQ ID NO:352) to generate peptide-PMOs (PPMOs). HEK293 cells were treated with PPMOs or controls for 24 h before total RNA was extracted using MagMAX total RNA 96 extraction kit method. PKD1 gene expression was assessed by digital droplet PCR (TaqMan; probe catalog number Hs00947394\_g1). PKD1 transcript expression was normalised to the housekeeper TATA-Binding Protein (TBP, TaqMan, probe catalog number Hs00427620\_m1) and changes calculated as fold-change compared to untreated cells.

[0129] As shown in FIG. 2, PPMOs corresponding to SEQ ID NOs:47 (+604+628), 334 (+611+635), and 335 (+615+639) respectively induced a dose-dependent increase of PKD1 mRNA greater than 1.1-fold compared to untreated cells. A non-target control (GTC CTR), predicted not to hybridize to human transcripts, was included as a sham treatment.

### Example 3: Effect on PC1 Protein Levels, in an ADPKD Patient Cell Line, of PPMOs that Mask the Sequence in the 3' UTR of PKD1 Targeted by the miR-17 miRNA Seed Sequence

[0130] PPMOs that demonstrated significant PKD1 mRNA upregulation, as per Example 2, were selected for their ability to upregulate Polycystin 1 (PC1) protein in an ADPKD patient cell line. The ADPKD patient cell line was established by immortalizing cells from a single proximal cortical tubule cyst taken from a patient with ADPKD, carrying the PKD1 heterozygous mutation (p.Q2556\*). PPMOs were dissolved in molecular grade H.sub.2O to generate 1 mM stock solutions. Stock solutions were diluted in media and dosed at 10  $\mu$ M with n=3 technical replicates per treatment condition. Untreated cells were treated with equivalent volume of media only. The cells were incubated for 5 days before the presence of Polycystin 1 (PC1) protein on the cell surface was measured via flow cytometry. The cells were incubated with an anti-rabbit Polycystin 1 antibody (ab74115, Abcam) followed by a goat anti-mouse antibody conjugated to Alexa Fluor 488 (A-11001, ThermoFisher). Comparison of the median fluorescence intensity (MFI) of each sample to untreated cells (UT) identified PMOs that increased expression of Polycystin 1, as illustrated in FIG. 3. PPMOs corresponding to SEQ ID NOs:47 (+604+628), 334 (+611+635), and 335 (+615+639) respectively induced an increase of Polycystin 1 protein greater than 1.1-fold compared to untreated cells.

**Example 4: Functional Validation of PPMOs in a Patient-Derived Primary Cell 3D Cyst Model**  
[0131] PPMOs that demonstrated significant Polycystin 1 protein upregulation, as per Example 3, were selected for functional validation in a patient-derived 3D cyst model. The model utilizes cyst cells extracted from kidneys donated by ADPKD patients. When grown ex vivo, these cells spontaneously form cysts in a 3D matrix and can be used to assess the functional effect of drug treatments on cyst growth. Immediately after seeding, patient cells were co-exposed to treatment. PPMOs were dissolved in molecular grade H.sub.2O and dosed at 1  $\mu$ M, 3  $\mu$ M, 10  $\mu$ M and 20  $\mu$ M with n=4 technical replicates per treatment condition. Control cells were treated with equivalent volume of H.sub.2O only. After 7 days of exposure, cultures were fixed and stained for actin cytoskeleton and nuclei. Growth and swelling of cysts were visualized by high-content microscopy imaging and images are analyzed using Ominer® image analysis software. FIG. 4 shows representative images of the size of 3D patient cyst after treated with 20  $\mu$ M PPMO with sequences corresponding to SEQ ID NOs:47 (+604+628), 334 (+611+635), and 335 (+615+639).

**Example 5: Quantification of Patient-Derived 3D Cyst Area and Cell Death to Determine the Therapeutic Index**

[0132] All images from the assay described in Example 4 were further analysed to determine the dose-dependent change of cyst area ( $\mu$ m.sup.2) and cell death (%) for PPMOs with sequences corresponding to SEQ ID NOs:47 (+604+628), 334 (+611+635), and 335 (+615+639). See FIG. 5.

[0133] The PPMO corresponding to SEQ ID NO:47 (+604+628) showed efficacy in inhibiting cyst growth in a dose dependent manner with no evidence of cytotoxicity at up to 20  $\mu$ M (FIGS. 5A and 5D). The PPMO corresponding to SEQ ID NO: 334 (+611+635) effectively inhibited cyst growth, however slight cytotoxicity was observed at 20  $\mu$ M treatment (FIGS. 5B and 5E). The PPMO corresponding to SEQ ID NO: 335 (+615+639) was found to be cytotoxic at 20  $\mu$ M. However, the PPMO demonstrated moderate inhibition of cyst growth at 10  $\mu$ M treatment in cyst models derived from ADPK patient (FIGS. 5C and 5F).

**Example 6: Optimization of PMO Sequences to Enhance PKD1 Upregulation Via Inhibition of miR-17 Binding to the 3'UTR of PKD1 Transcripts**

[0134] PMOs (SEQ ID NOs:353-361) were designed to microwalk SEQ ID NO:47 with some modifications, e.g., length shortening and engineered-base mismatch. Microwalked PMOs were conjugated with a cell-penetrating peptide (SEQ ID NO:352) to generate PPMOs and tested for their efficacy in PKD1 upregulation in HEK293 cells. HEK293 cells were treated with PPMOs or controls for 24 hours. Subsequently, total RNA was extracted using the MagMAX Total RNA 96 Extraction Kit method. The expression of PKD1 transcript was assessed by digital droplet PCR (TaqMan; probe catalog number Hs00947394\_g1). PKD1 transcript expression was normalized to the housekeepers TATA-Binding Protein (TBP, TaqMan; probe catalog number Hs00427620\_m1) and DexH-Box Helicase 57 (DHX57, ThermoFisher; Assay ID Hs00376574\_m1), and changes were calculated as fold-change compared to untreated cells. The results in FIG. 6 demonstrated a slight PKD1 upregulation (1.1-fold) when treated with PPMOs (+604+626)MM16, (+604+628)MM16, (+604+626), (+604+623), (+604+628) and (+604+627).

**Example 7: Screening of PPMOs to Enhance PKD1 Upregulation Via Inhibition of miR-200 Binding to the 3'UTR of PKD1 Transcripts**

[0135] PMOs with sequences corresponding to SEQ ID NOs:14, 73, 75, 336-337, were designed to target binding the 3' UTR of the PKD1 mRNA to prevent binding of miR-200. Designed PMOs were conjugated to a cell penetrating peptide (SEQ ID NO:352) to generate peptide-PMOs (PPMOs) and tested for their efficacy in HEK293 cells. Following for 24 h of PPMO treatment, total RNA was extracted using the MagMAX total RNA 96 extraction kit method. PKD1 gene expression was assessed by digital droplet PCR (TaqMan; probe catalog number Hs00947394\_g1). PKD1 transcript expression was normalised to the housekeeper TATA-Binding Protein (TBP, TaqMan, probe catalog number Hs00427620\_m1) and changes calculated as fold-change compared to untreated cells. As shown in FIG. 7, PPMOs corresponding to SEQ ID NOs:17 (+505+529), 73

(+515+539), 336 (+677+701), and 337 (+688+712) slightly increased the expression of PKD1 mRNA (1.1-fold) compared to untreated cells.

#### Example 8: PC1 Upregulation Assessment for PPMO-Mediated Inhibition of miR-17 Binding in HEK293 Cells

[0136] PPMOs with the sequences corresponding to SEQ ID NOs:47 (+604+628) and 334 (+611+635) were incubated to HEK293 for 5 days. The upregulation level of PC1 was assessed using western blot assay. At day 5 post-treatment, protein was extracted using RIPA buffer supplemented with 2% protease inhibitor cocktail and 2× PhosSTOP. Protein lysates were cleared, total protein quantitated using BCA protein kit. Samples were run on NuPAGE 3-8% Tris Acetate protein gels. Protein was transferred to nitrocellulose membrane by wet transfer. The membrane was stained for total protein and mouse anti-PC1 (Santa-Cruz, cat no sc130554) primary antibody followed by anti-mouse (IRDye® 800CW preabsorbed) secondary antibody. Blots were imaged on an Odyssey Imager, quantitative analysis was performed using Image Studio Ver 5.5 software (shown in FIG. 8). The raw fluorescence signal for PC1 was first normalized to the raw fluorescence signal of loading control (total protein) and then expressed as fold-change relative to UT. As shown in FIG. 9, both PPMOs (+604+628) and (+611+635) induced dose dependent increase in PC1 protein up to 1.64 fold compared to UT up to day 5. A non-targeting control, predicted not to hybridize to human transcripts, was included as a negative control.

#### Example 9: Time-Course and Dose-Range Analysis of PPMO in Inducing PC1 Protein in Cell Line Derived from an ADPKD Patient

[0137] WT9-7 cell line harboring a PKD1 nonsense mutation (p.Q2556\*) was incubated with a PPMO (+604+628) corresponding to SEQ ID NO:47 at concentrations of 30 PM, 60 PM, 90 μM and 120 μM and incubated for 2, 3 and 5 days. PC1 protein levels were assessed to evaluate the efficacy of the PPMO. Protein extraction was performed using RIPA buffer supplemented with 2% protease inhibitor cocktail and 2× PhosSTOP. Protein lysates were cleared, total protein quantitated using BCA protein kit. Samples were run on NuPAGE 3-8% Tris Acetate protein gels. Protein was transferred to nitrocellulose membrane by wet transfer. The membrane was stained for total protein and mouse anti-PC1 (Santa-Cruz, Cat No. sc130554) primary antibody followed by anti-mouse (IRDye® 800CW preabsorbed) secondary antibody. Blots were imaged on an Odyssey Imager, quantitative analysis was performed using Image Studio Ver 5.5 software. The raw fluorescence signal for PC1 was first normalized to the raw fluorescence signal of loading control (total protein) and then expressed as fold-change relative to UT. Western blot images demonstrated PC1 expression in FIG. 10. PPMO (+604+628)-SEQ ID NO:47 induced a dose-dependent increase in PC1 protein levels, up to 1.36-fold compared to UT. The maximum PC1 upregulation was observed on day 2 and appeared to plateau at 90 μM.

[0138] It will be appreciated by persons skilled in the art that numerous variations and/or modifications may be made to the invention as shown in the specific examples without departing from the spirit or scope of the invention as broadly described. The present examples are, therefore, to be considered in all respects as illustrative and not restrictive.

[0139] All publications cited herein are hereby incorporated by reference in their entirety. Where reference is made to a URL or other such identifier or address, it is understood that such identifiers can change and particular information on the internet can come and go, but equivalent information can be found by searching the internet. Reference thereto evidences the availability and public dissemination of such information.

[0140] Any discussion of documents, acts, materials, devices, articles or the like that has been included in the present specification is solely for the purpose of providing a context for the present invention. It is not to be taken as an admission that any or all of these matters form part of the prior art base or were common general knowledge in the field relevant to the present invention as it existed before the priority date of each claim of this application.

#### REFERENCES

[0141] de Carvalho Ribeiro et al., 2020, Stem Cells International, doi.org/10.1155/2020/8894590. [0142] Dugal-Tessier et al., (2021), *J Clin Med.*, 10(4):838. [0143] Osafune et al., 2021, Clinical and Experimental Nephrology, 25(6):574-584. [0144] Prakash et al., 2019, Nucleic Acids Research, 47(12):6029-6044. [0145] Sharma et al., 2019, Methods Cell Biol., doi.org/10.1016/bs.mcb.2019.05.008

TABLE-US-00001 APPENDIX 1 Sequences and SEQ ID NOS SEQ ID NO: 1  
Human PKD1 transcript 3' UTR (PKD-201 canonical transcript  
ENST00000262304.9)

TCCTCCTTCCTGGCGGGGGTGGGCCGTGGAGTCGGAGTGGACACCGCTCAGTATT  
ACTTTCTGCCGCTGTCAAGGCCGAGGGCCAGGCAGAATGGCTGCACGTAGGTTC  
CCCAGAGAGCAGGCAGGGGCATCTGTCTGTCTGTGGGCTTCAGCACTTTAAAGA  
GGCTGTGTGGCCAACCAGGACCCAGGGTCCCCTCCCCAGCTCCCTTGGGAAGGA  
CACAGCAGTATTGGACGGTTTCTAGCCTCTGAGATGCTAATTTATTTCCCCGAGT  
CCTCAGGTACAGCGGGCTGTGCCCGGCCCCACCCCCTGGGCAGATGTCCCCCACT  
GCTAAGGCTGCTGGCTTCAGGGAGGGTTAGCCTGCACCGCCGCCACCCTGCCCCCT  
AAGTTATTACCTCTCCAGTTCCTACCGTACTCCCTGCACCGTCTCACTGTGTGTCT  
CGTGTCAAGTAATTTATATGGTGTAAATGTGTATATTTTGTATGTCACTATTTT  
CACTAGGGCTGAGGGGCCTGCGCCCAGAGCTGGCCTCCCCCAACACCTGCTGCG  
CTTGGTAGGTGTGGTGGCGTTATGGCAGCCCGGCTGCTGCTTGGATGCGAGCTTG  
GCCTTGGGCGGTGCTGGGGGCACAGCTGTCTGCCAGGCACTCTCATCACCCAG  
AGGCCTTGTCATCCTCCCTTGCCCCAGGCCAGGTAGCAAGAGAGCAGCGCCCAG  
GCCTGCTGGCATCAGGTCTGGGCAAGTAGCAGGACTAGGCATGTCAGAGGACCC  
CAGGGTGGTTAGAGGAAAAGACTCCTCCTGGGGGCTGGCTCCCAGGGTGGAGGA  
AGGTGACTGTGTGTGTGTGTGTGTGTGCGCGCGCGCACGCGCGAGTGTGCTGTATGG  
CCCAGGCAGCCTCAAGGCCCTCGGAGCTGGCTGTGCCTGCTTCTGTGTACCACTT  
CTGTGGGCATGGCCGCTTCTAGAGCCTCGACACCCCCCAACCCCCGCACCAAGC  
AGACAAAGTCAATAAAAGAGCTGTCTGACTGCAA

TABLE-US-00002 TABLE 1 Exemplary ASO or AR Sequences Targeting PKD1  
3' UTR Sequences SEQ ID NO ASO\_Seq ASO\_Name 2

|                            |      |     |                |    |
|----------------------------|------|-----|----------------|----|
| GGCCCACCCCCGCCAGGAAGGAGGA  | PKD1 | H46 | 3UTR(+469+493) | 3  |
| CACGGCCCCACCCCCGCCAGGAAGGA | PKD1 | H46 | 3UTR(+472+496) | 4  |
| CTCCACGGCCCCACCCCCGCCAGGAA | PKD1 | H46 | 3UTR(+475+499) | 5  |
| CGACTCCACGGCCCCACCCCCGCCAG | PKD1 | H46 | 3UTR(+478+502) | 6  |
| CTCCGACTCCACGGCCCCACCCCCGC | PKD1 | H46 | 3UTR(+481+505) | 7  |
| CCACTCCGACTCCACGGCCCCACCCC | PKD1 | H46 | 3UTR(+484+508) | 8  |
| TGTCCACTCCGACTCCACGGCCCCAC | PKD1 | H46 | 3UTR(+487+511) | 9  |
| CGGTGTCCACTCCGACTCCACGGCC  | PKD1 | H46 | 3UTR(+490+514) | 10 |
| GAGCGGTGTCCACTCCGACTCCACG  | PKD1 | H46 | 3UTR(+493+517) | 11 |
| ACTGAGCGGTGTCCACTCCGACTCC  | PKD1 | H46 | 3UTR(+496+520) | 12 |
| AATACTGAGCGGTGTCCACTCCGAC  | PKD1 | H46 | 3UTR(+499+523) | 13 |
| AGTAATACTGAGCGGTGTCCACTCC  | PKD1 | H46 | 3UTR(+502+526) | 14 |
| GAAAGTAATACTGAGCGGTGTCCAC  | PKD1 | H46 | 3UTR(+505+529) | 15 |
| GCAGAAAGTAATACTGAGCGGTGTC  | PKD1 | H46 | 3UTR(+508+532) | 16 |
| GCGGCAGAAAGTAATACTGAGCGGT  | PKD1 | H46 | 3UTR(+511+535) | 17 |
| ACAGCGGCAGAAAGTAATACTGAGC  | PKD1 | H46 | 3UTR(+514+538) | 18 |
| TTGACAGCGGCAGAAAGTAATACTG  | PKD1 | H46 | 3UTR(+517+541) | 19 |
| GCCTTGACAGCGGCAGAAAGTAATA  | PKD1 | H46 | 3UTR(+520+544) | 20 |
| TCGGCCTTGACAGCGGCAGAAAGTA  | PKD1 | H46 | 3UTR(+523+547) | 21 |
| CCCTCGGCCTTGACAGCGGCAGAAA  | PKD1 | H46 | 3UTR(+526+550) | 22 |
| TGGCCCTCGGCCTTGACAGCGGCAG  | PKD1 | H46 | 3UTR(+529+553) | 23 |

|                                 |     |                |    |
|---------------------------------|-----|----------------|----|
| GGCTGGCCCTCGGACCTTGACAGCGG PKD1 | H46 | 3UTR(+532+556) | 24 |
| TCTGCCTGGCCCTCGGCCTTGACAG PKD1  | H46 | 3UTR(+535+559) | 25 |
| CATTCTGCCTGGCCCTCGGCCTTGA PKD1  | H46 | 3UTR(+538+562) | 26 |
| AGCCATTCTGCCTGGCCCTCGGCCT PKD1  | H46 | 3UTR(+541+565) | 27 |
| TGCAGCCATTCTGCCTGGCCCTCGG PKD1  | H46 | 3UTR(+544+568) | 28 |
| ACGTGCAGCCATTCTGCCTGGCCCT PKD1  | H46 | 3UTR(+547+571) | 29 |
| CCTACGTGCAGCCATTCTGCCTGGC PKD1  | H46 | 3UTR(+550+574) | 30 |
| GAACCTACGTGCAGCCATTCTGCCT PKD1  | H46 | 3UTR(+553+577) | 31 |
| GGGGAACCTACGTGCAGCCATTCTG PKD1  | H46 | 3UTR(+556+580) | 32 |
| TCTGGGGAACCTACGTGCAGCCATT PKD1  | H46 | 3UTR(+559+583) | 33 |
| CTCTCTGGGGAACCTACGTGCAGCC PKD1  | H46 | 3UTR(+562+586) | 34 |
| CTGCTCTCTGGGGAACCTACGTGCA PKD1  | H46 | 3UTR(+565+589) | 35 |
| TGCCTGCTCTCTGGGGAACCTACGT PKD1  | H46 | 3UTR(+568+592) | 36 |
| CCCTGCCTGCTCTCTGGGGAACCTA PKD1  | H46 | 3UTR(+571+595) | 37 |
| TGCCCCTGCCTGCTCTCTGGGGAAC PKD1  | H46 | 3UTR(+574+598) | 38 |
| AGATGCCCCTGCCTGCTCTCTGGGG PKD1  | H46 | 3UTR(+577+601) | 39 |
| GACAGATGCCCCTGCCTGCTCTCTG PKD1  | H46 | 3UTR(+580+604) | 40 |
| ACAGACAGATGCCCCTGCCTGCTCT PKD1  | H46 | 3UTR(+583+607) | 41 |
| CAGACAGACAGATGCCCCTGCCTGC PKD1  | H46 | 3UTR(+586+610) | 42 |
| CCACAGACAGACAGATGCCCCTGCC PKD1  | H46 | 3UTR(+589+613) | 43 |
| AGCCCACAGACAGACAGATGCCCCT PKD1  | H46 | 3UTR(+592+616) | 44 |
| TGAAGCCCACAGACAGACAGATGCC PKD1  | H46 | 3UTR(+595+619) | 45 |
| TGCTGAAGCCCACAGACAGACAGAT PKD1  | H46 | 3UTR(+598+622) | 46 |
| AAGTGCTGAAGCCCACAGACAGACA PKD1  | H46 | 3UTR(+601+625) | 47 |
| TTAAAGTGCTGAAGCCCACAGACAG PKD1  | H46 | 3UTR(+604+628) | 48 |
| TCTTTAAAGTGCTGAAGCCCACAGA PKD1  | H46 | 3UTR(+607+631) | 49 |
| GCCTCTTTAAAGTGCTGAAGCCCAC PKD1  | H46 | 3UTR(+610+634) | 50 |
| ACAGCCTCTTTAAAGTGCTGAAGCC PKD1  | H46 | 3UTR(+613+637) | 51 |
| CACACAGCCTCTTTAAAGTGCTGAA PKD1  | H46 | 3UTR(+616+640) | 52 |
| GGCCACACAGCCTCTTTAAAGTGCT PKD1  | H46 | 3UTR(+619+643) | 53 |
| GTTGGCCACACAGCCTCTTTAAAGT PKD1  | H46 | 3UTR(+622+646) | 54 |
| CTGGTTGGCCACACAGCCTCTTTAA PKD1  | H46 | 3UTR(+625+649) | 55 |
| GTCCTGGTTGGCCACACAGCCTCTT PKD1  | H46 | 3UTR(+628+652) | 56 |
| TGGGTCCTGGTTGGCCACACAGCCT PKD1  | H46 | 3UTR(+631+655) | 57 |
| CCCTGGGTCCTGGTTGGCCACACAG PKD1  | H46 | 3UTR(+634+658) | 58 |
| GGACCCTGGGTCCTGGTTGGCCACA PKD1  | H46 | 3UTR(+637+661) | 59 |
| AGGGGACCCTGGGTCCTGGTTGGCC PKD1  | H46 | 3UTR(+640+664) | 60 |
| GGGAGGGGACCCTGGGTCCTGGTTG PKD1  | H46 | 3UTR(+643+667) | 61 |
| CTGGGGAGGGGACCCTGGGTCCTGG PKD1  | H46 | 3UTR(+646+670) | 62 |
| GAGCTGGGGAGGGGACCCTGGGTCC PKD1  | H46 | 3UTR(+649+673) | 63 |
| AGGGAGCTGGGGAGGGGACCCTGGG PKD1  | H46 | 3UTR(+652+676) | 64 |
| CCAAGGGAGCTGGGGAGGGGACCCT PKD1  | H46 | 3UTR(+655+679) | 65 |
| TTCCCAAGGGAGCTGGGGAGGGGAC PKD1  | H46 | 3UTR(+658+682) | 66 |
| TCCTTCCCAAGGGAGCTGGGGAGGG PKD1  | H46 | 3UTR(+661+685) | 67 |
| GTGTCCTTCCCAAGGGAGCTGGGGA PKD1  | H46 | 3UTR(+664+688) | 68 |
| GCTGTGTCCTTCCCAAGGGAGCTGG PKD1  | H46 | 3UTR(+667+691) | 69 |
| ACTGCTGTGTCCTTCCCAAGGGAGC PKD1  | H46 | 3UTR(+670+694) | 70 |
| AATACTGCTGTGTCCTTCCCAAGGG PKD1  | H46 | 3UTR(+673+697) | 71 |
| TCCAATACTGCTGTGTCCTTCCCAA PKD1  | H46 | 3UTR(+676+700) | 72 |
| CCGTCCAATACTGCTGTGTCCTTCC PKD1  | H46 | 3UTR(+679+703) | 73 |

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| AAACCGTCCCACTGCTGTGCTGCT PKD1    | H46 | 3UTR(+682+706) 74  |
| TAGAAACCGTCCAATACTGCTGTGT PKD1   | H46 | 3UTR(+685+709) 75  |
| GGCTAGAAACCGTCCAATACTGCTG PKD1   | H46 | 3UTR(+688+712) 76  |
| AGAGGCTAGAAACCGTCCAATACTG PKD1   | H46 | 3UTR(+691+715) 77  |
| CTCAGAGGCTAGAAACCGTCCAATA PKD1   | H46 | 3UTR(+694+718) 78  |
| CATCTCAGAGGCTAGAAACCGTCCA PKD1   | H46 | 3UTR(+697+721) 79  |
| TAGCATCTCAGAGGCTAGAAACCGT PKD1   | H46 | 3UTR(+700+724) 80  |
| AATTAGCATCTCAGAGGCTAGAAAC PKD1   | H46 | 3UTR(+703+727) 81  |
| ATAAATTAGCATCTCAGAGGCTAGA PKD1   | H46 | 3UTR(+706+730) 82  |
| GAAATAAATTAGCATCTCAGAGGCT PKD1   | H46 | 3UTR(+709+733) 83  |
| GGGGAAATAAATTAGCATCTCAGAG PKD1   | H46 | 3UTR(+712+736) 84  |
| CTCGGGGAAATAAATTAGCATCTCA PKD1   | H46 | 3UTR(+715+739) 85  |
| GGA CT CGGGGAAATAAATTAGCATC PKD1 | H46 | 3UTR(+718+742) 86  |
| TGAGGACTCGGGGAAATAAATTAGC PKD1   | H46 | 3UTR(+721+745) 87  |
| ACCTGAGGACTCGGGGAAATAAATT PKD1   | H46 | 3UTR(+724+748) 88  |
| TGTACCTGAGGACTCGGGGAAATAA PKD1   | H46 | 3UTR(+727+751) 89  |
| CGCTGTACCTGAGGACTCGGGGAAA PKD1   | H46 | 3UTR(+730+754) 90  |
| GCCCGCTGTACCTGAGGACTCGGGG PKD1   | H46 | 3UTR(+733+757) 91  |
| ACAGCCCGCTGTACCTGAGGACTCG PKD1   | H46 | 3UTR(+736+760) 92  |
| GGCACAGCCCGCTGTACCTGAGGAC PKD1   | H46 | 3UTR(+739+763) 93  |
| CCGGGCACAGCCCGCTGTACCTGAG PKD1   | H46 | 3UTR(+742+766) 94  |
| GGGCCGGGCACAGCCCGCTGTACCT PKD1   | H46 | 3UTR(+745+769) 95  |
| GTGGGGCCGGGCACAGCCCGCTGTA PKD1   | H46 | 3UTR(+748+772) 96  |
| GGGGTGGGGCCGGGCACAGCCCGCT PKD1   | H46 | 3UTR(+751+775) 97  |
| CAGGGGGTGGGGCCGGGCACAGCCC PKD1   | H46 | 3UTR(+754+778) 98  |
| GCCCAGGGGGTGGGGCCGGGCACAG PKD1   | H46 | 3UTR(+757+781) 99  |
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| GGGACATCTGCCCAGGGGGTGGGGC PKD1   | H46 | 3UTR(+766+790) 102 |
| TGGGGGACATCTGCCCAGGGGGTGG PKD1   | H46 | 3UTR(+769+793) 103 |
| CAGTGGGGGACATCTGCCCAGGGGG PKD1   | H46 | 3UTR(+772+796) 104 |
| TAGCAGTGGGGGACATCTGCCCAGG PKD1   | H46 | 3UTR(+775+799) 105 |
| CCTTAGCAGTGGGGGACATCTGCCC PKD1   | H46 | 3UTR(+778+802) 106 |
| CAGCCTTAGCAGTGGGGGACATCTG PKD1   | H46 | 3UTR(+781+805) 107 |
| CAGCAGCCTTAGCAGTGGGGGACAT PKD1   | H46 | 3UTR(+784+808) 108 |
| AGCCAGCAGCCTTAGCAGTGGGGGA PKD1   | H46 | 3UTR(+787+811) 109 |
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| CCTCCCTGAAGCCAGCAGCCTTAGC PKD1   | H46 | 3UTR(+796+820) 112 |
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| GGCAGGGTGGCGGCGGTGCAGGCTA PKD1   | H46 | 3UTR(+823+847) 121 |
| AGGGGCAGGGTGGCGGCGGTGCAGG PKD1   | H46 | 3UTR(+826+850) 122 |
| CTTAGGGGCAGGGTGGCGGCGGTGC PKD1   | H46 | 3UTR(+829+853) 123 |



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| TAATAACTTAGGGGCAGGGTGGCGG PKD1  | H46 | 3UTR(+835+859)  | 125 |
| AGGTAATAACTTAGGGGCAGGGTGG PKD1  | H46 | 3UTR(+838+862)  | 126 |
| GAGAGGTAATAACTTAGGGGCAGGG PKD1  | H46 | 3UTR(+841+865)  | 127 |
| CTGGAGAGGTAATAACTTAGGGGGCA PKD1 | H46 | 3UTR(+844+868)  | 128 |
| GAAGTGGAGAGGTAATAACTTAGGG PKD1  | H46 | 3UTR(+847+871)  | 129 |
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| GAGACGGTGCAGGGAGTACGGTAGG PKD1  | H46 | 3UTR(+871+895)  | 137 |
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| TGACACGAGACACACAGTGAGACGG PKD1  | H46 | 3UTR(+889+913)  | 143 |
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| TAGTGAAAATAGTGACATACAAAAA PKD1  | H46 | 3UTR(+943+967)  | 161 |
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| GCCCCTCAGCCCTAGTGAAAATAGT PKD1  | H46 | 3UTR(+955+979)  | 165 |
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| GGAGGCCAGCTCTGGGCGCAGGCC PKD1   | H46 | 3UTR(+976+1000) | 172 |
| GGGGGAGGCCAGCTCTGGGCGCAGG PKD1  | H46 | 3UTR(+979+1003) | 173 |

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| GGTGTGGGGGAGGCCAGCTCTGGG PKD1   | H46 | 3UTR(+985+1009) 175  |
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| GCAGCAGGTGTTGGGGGAGGCCAGC PKD1  | H46 | 3UTR(+991+1015) 177  |
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| ATAACGCCACCACACCTACCAAGCG PKD1  | H46 | 3UTR(+1015+1039) 185 |
| GCCATAACGCCACCACACCTACCAA PKD1  | H46 | 3UTR(+1018+1042) 186 |
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| CGCGCGCGCACACACACACACACAC PKD1   | H46 | 3UTR(+1297+1321) 279 |
| GTGCGCGCGCGCGCACACACACACA PKD1   | H46 | 3UTR(+1300+1324) 280 |
| CGCGTGTGCGCGCGCGCACACACAC PKD1   | H46 | 3UTR(+1303+1327) 281 |
| TCGCGCGTGTGCGCGCGCGCACACA PKD1   | H46 | 3UTR(+1306+1330) 282 |
| CACTCGCGCGTGTGCGCGCGCGCACAC PKD1 | H46 | 3UTR(+1309+1333) 283 |
| GCACACTCGCGCGTGTGCGCGCGCGCA PKD1 | H46 | 3UTR(+1312+1336) 284 |
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| ATGCCCACAGAAGTGGTACACAGAA PKD1   | H46 | 3UTR(+1384+1408) 308 |
| GCCATGCCCACAGAAGTGGTACACA PKD1   | H46 | 3UTR(+1387+1411) 309 |
| GCGGCCATGCCCACAGAAGTGGTAC PKD1   | H46 | 3UTR(+1390+1414) 310 |
| GAAGCGGCCATGCCCACAGAAGTGG PKD1   | H46 | 3UTR(+1393+1417) 311 |
| CTAGAAGCGGCCATGCCCACAGAAG PKD1   | H46 | 3UTR(+1396+1420) 312 |
| GCTCTAGAAGCGGCCATGCCCACAG PKD1   | H46 | 3UTR(+1399+1423) 313 |
| GAGGCTCTAGAAGCGGCCATGCCCCA PKD1  | H46 | 3UTR(+1402+1426) 314 |
| GTCGAGGCTCTAGAAGCGGCCATGC PKD1   | H46 | 3UTR(+1405+1429) 315 |
| GGTGTGAGGCTCTAGAAGCGGCCA PKD1    | H46 | 3UTR(+1408+1432) 316 |
| GGGGGTGTCGAGGCTCTAGAAGCGG PKD1   | H46 | 3UTR(+1411+1435) 317 |
| TGGGGGGGTGTCGAGGCTCTAGAAG PKD1   | H46 | 3UTR(+1414+1438) 318 |
| GGTTGGGGGGGTGTCGAGGCTCTAG PKD1   | H46 | 3UTR(+1417+1441) 319 |
| GGGGGTTGGGGGGGTGTCGAGGCTC PKD1   | H46 | 3UTR(+1420+1444) 320 |
| TGCGGGGGTTGGGGGGGTGTCGAGG PKD1   | H46 | 3UTR(+1423+1447) 321 |
| TGGTGCGGGGGTTGGGGGGGTGTCG PKD1   | H46 | 3UTR(+1426+1450) 322 |
| GCTTGGTGCGGGGGTTGGGGGGGTG PKD1   | H46 | 3UTR(+1429+1453) 323 |

|                                    |                    |                         |     |
|------------------------------------|--------------------|-------------------------|-----|
| TCGTGCTTGGTGCGGGGGGTTGGGGG PKD1    | H46                | 3UTR(+1432+1456)        | 324 |
| TTGTCTGCTTGGTGCGGGGGGTTGGG PKD1    | H46                | 3UTR(+1435+1459)        | 325 |
| ACTTTGTCTGCTTGGTGCGGGGGGTT PKD1    | H46                | 3UTR(+1438+1462)        | 326 |
| TTGACTTTGTCTGCTTGGTGCGGGG PKD1     | H46                | 3UTR(+1441+1465)        | 327 |
| TTATTGACTTTGTCTGCTTGGTGCG PKD1     | H46                | 3UTR(+1444+1468)        | 328 |
| CTTTTATTGACTTTGTCTGCTTGGT PKD1     | H46                | 3UTR(+1447+1471)        | 329 |
| GCTCTTTTATTGACTTTGTCTGCTT PKD1     | H46                | 3UTR(+1450+1474)        | 330 |
| ACAGCTCTTTTATTGACTTTGTCTG PKD1     | H46                | 3UTR(+1453+1477)        | 331 |
| CAGACAGCTCTTTTATTGACTTTGT PKD1     | H46                | 3UTR(+1456+1480)        | 332 |
| AGTCAGACAGCTCTTTTATTGACTT PKD1     | H46                | 3UTR(+1459+1483)        | 333 |
| TGCAGTCAGACAGCTCTTTTATTGA PKD1     | H46                | 3UTR(+1462+1486)        | 334 |
| AGCCTCTTTAAAGTGCTGAAGCCCA PKD1     | H46                | 3UTR(+611+635)          | 335 |
| ACACAGCCTCTTTAAAGTGCTGAAG PKD1     | H46                | 3UTR(+615+639)          | 336 |
| GTCCAATACTGCTGTGTCCTTCCCA PKD1     | H46                | 3UTR(+677+701)          | 337 |
| GACAGCGGCAGAAAGTAATACTGAG PKD1     | H46                | 3UTR(+515+539)          | 338 |
| CTCTTTAAAGTGCTGAAGCCCACAG PKD1     | H46                | 3UTR(+608+632)          | 339 |
| CAGCCTCTTTAAAGTGCTGAAGCCC PKD1     | H46                | 3UTR(+612+636)          | 340 |
| CACAGCCTCTTTAAAGTGCTGAAGC PKD1     | H46                | 3UTR(+614+638)          | 341 |
| TTAAAGTGCTGAAGCCCACATACAG PKD1     | H46                | 3UTR(+604+628)MM, G > T | 342 |
| TCTTTAAAGTGCTGAAGCCCACATA PKD1     | H46                | 3UTR(+607+631)MM, G > T | 343 |
| CTCTTTAAAGTGCTGAAGCACACAG PKD1     | H46                | 3UTR(+608+632)MM, C > A | 344 |
| GCCTCCTTAAAGTGCTGAAGCCCAC PKD1     | H46                | 3UTR(+610+634)MM, T > C | 345 |
| AGCCTCTATAAAGTGCTGAAGCCCA PKD1     | H46                | 3UTR(+611+635)MM, T > A | 346 |
| CAGCCTCTATAAAGTGCTGAAGCCC PKD1     | H46                | 3UTR(+612+636)MM, T > A | 347 |
| ACAGCCTCTTTAAAGTGCTGAATCC PKD1     | H46                | 3UTR(+613+637)MM, G > T | 348 |
| CACAGCCTCTTTAAAGTGCTGAATC PKD1     | H46                | 3UTR(+614+638)MM, G > T | 349 |
| ACACAACCTCTTTAAAGTGCTGAAG PKD1     | H46                | 3UTR(+615+639)MM, G > A | 350 |
| CACATAGCCTCTTTAAAGTGCTGAA PKD1     | H46                | 3UTR(+616+640)MM, C > T | 351 |
| GGCCACACAGTCTCTTTAAAGTGCT PKD1     | H46                | 3UTR(+619+643)MM, C > T | 353 |
| TTAAAGTGCTGAAGCCCAC PKD1_H46       | 3UTR(+604+622)     |                         | 354 |
| TTAAAGTGCTGAAGCCCACAG PKD1_H46     | 3UTR(+604+624)     |                         | 355 |
| TTAAAGTGCTGAAGCCCACAGAC PKD1_H46   | 3UTR(+604+626)     |                         | 356 |
| TTAAAGTGCTGAAGCTCACAGAC PKD1_H46   | 3UTR(+604+626)MM16 |                         | 357 |
| TTAAAGTGCTGAAGCTCACAGACAG PKD1_H46 | 3UTR(+604+628)MM16 |                         | 358 |
| TTAAAGTGCTGAAGCCCACA PKD1_H46      | 3UTR(+604+623)     |                         | 359 |
| TTAAAGTGCTGAAGCCCACAGA PKD1_H46    | 3UTR(+604+625)     |                         | 360 |
| TTAAAGTGCTGAAGCCCACAGACA PKD1_H46  | 3UTR(+604+627)     |                         | 361 |
| TTAAAGTACTGAAGCCCACAGACAG PKD1_H46 | 3UTR(+604+628)MM8  |                         | 362 |
| GTGCTGAAGCCCACAGACAGACAG PKD1_H46  | 3UTR(+603+626)     |                         |     |

TABLE-US-00003 Amino acid sequence of CPP. SEQ ID NO: 352  
RRSRTARAGRPRNSSRPSAPR

## Claims

1-55. (canceled)

56. An antisense oligonucleotide, wherein the nucleotide sequence of the antisense oligonucleotide consists of SEQ ID NO:47, wherein the antisense oligonucleotide is a phosphorodiamidate morpholino oligonucleotide.

57. A pharmaceutical composition comprising the antisense oligonucleotide according to claim 56, and a pharmaceutically acceptable excipient.

- 58.** A method for treating autosomal dominant polycystic kidney disease (ADPKD), the method comprising administering to a subject in need thereof a therapeutically effective amount of the pharmaceutical composition according to claim 57.
- 59.** The method according to claim 58, wherein the subject is a human subject.
- 60.** The antisense oligonucleotide according to claim 56, wherein the antisense oligonucleotide further comprises a cell-penetrating peptide (CPP) covalently linked to the antisense oligonucleotide.
- 61.** The antisense oligonucleotide according to claim 60, wherein the amino acid sequence of the CPP comprises SEQ ID NO:352.
- 62.** The antisense oligonucleotide according to claim 61, wherein the amino acid sequence of the CPP consists of SEQ ID NO:352.
- 63.** The antisense oligonucleotide according to claim 61, wherein any amino acid other than glycine is a D amino acid.
- 64.** The antisense oligonucleotide according to claim 60, wherein the CPP is linked to the 5' end of the antisense oligonucleotide.
- 65.** A pharmaceutical composition comprising the antisense oligonucleotide according to claim 60, and a pharmaceutically acceptable excipient.
- 66.** A method for treating autosomal dominant polycystic kidney disease (ADPKD), the method comprising administering to a subject in need thereof a therapeutically effective amount of the pharmaceutical composition according to claim 65.
- 67.** The method according to claim 66, wherein the subject is a human subject.
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